

Final Project

NTSC Video Processing System

ENSC 425 – Electronic System Design

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Introduction

The purpose of this project was to design and implement a circuit capable of decoding a standard NTSC composite video signal and displaying the resulting image on the cathode ray tube (CRT) of an analog oscilloscope, using the X-, Y-, and Z-axis inputs for horizontal deflection, vertical deflection, and beam-intensity control, respectively. The system separates the horizontal and vertical synchronization signals to generate the required sweep waveforms while processing the luminance signal separately for image display. The overall design consists of input processing, sync separation, sweep generation, blanking, and luminance processing subcircuits, which were individually designed, tested, and then integrated into the final system.

High-Level Specification

Input Signal:

Standard NTSC video, RCA connector with adapter.

Three Output Signals:

V_{H-ramp} : Applied to X-Channel of oscilloscope.

V_{V-ramp} : Applied to Y-Channel of oscilloscope.

V_{Lumin} : Applied to Z-Channel of oscilloscope.

Controls:

Two trim-pots to adjust Contrast and Brightness.

Timing:

The video timing should correctly display odd/even fields with accurate interlace timing.

Blanking:

The luminance output signal should be blanked during both the horizontal and vertical porches.

Overscan:

When the input signal is not present, both ramp signals should be free-running and continue to scan. Obviously sync extraction will not be possible so a reasonable amount of overscan should be specified.

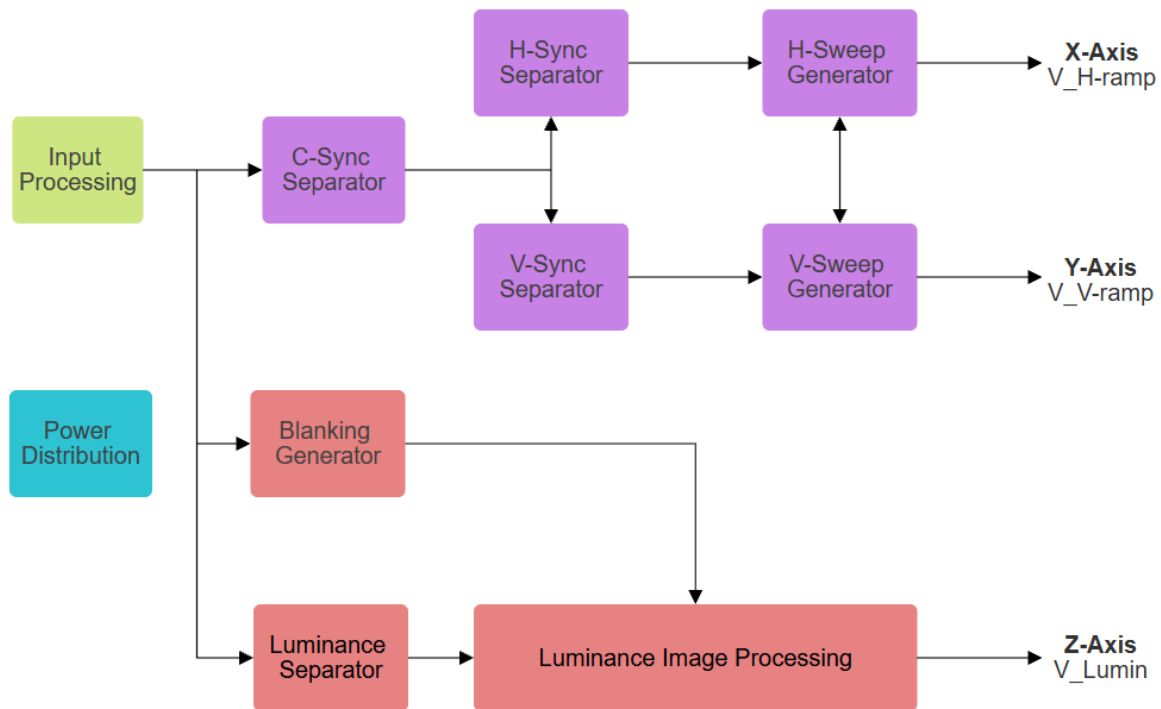


Figure 1: High-Level Block Diagram of the NTSC Video Processing System

Subcircuit Interface Specifications

Power Supply Levels

Function

- Provide the required supply voltages for each circuit section.
- Generate a 5 V supply from the available ± 12 V supplies.

Requirements

- Provide stable +12 V and -12 V supply rails.
- Provide a regulated +5 V supply for ICs requiring lower supply voltages.
- Keep all supply voltages within component ratings.

Inputs

- +12 V and -12 V laboratory power supplies.

Outputs

- +12 V and -12 V rails.
- +5 V supply rail.

Input Buffering, Clamping

Function:

- Isolates the input signal and produces a stable signal for further processing.

Requirements:

- Output should be independent of average frame luminance.
- Preserve the required video bandwidth with minimal distortion.
- Provide sufficient isolation from downstream loading.

Inputs:

- Standard NTSC composite video signal.

Outputs:

- Buffered and clamped NTSC composite video signal.

Composite Sync Separator

Function:

- Extract only the synchronization information from the NTSC signal.

Requirements:

- All sync tips must be present with minimal delay and reasonably sharp edges.

Inputs:

- Buffered and clamped NTSC composite video signal.

Outputs:

- Composite sync signal containing the NTSC synchronization pulses.

Horizontal Sync Separator

Function:

- Produce horizontal sync pulses only.

Requirements:

- Horizontal sync pulses should occur only at the line frequency.
- Output pulses should be approximately 4 μ s, which are typical for H-sync tips.
Equalization pulses during the vertical retrace period must be correctly processed, producing a continuous stream of H-sync pulses at the line frequency.

Inputs:

- Composite sync signal.

Outputs:

- Horizontal sync pulse signal.

Vertical Sync Separator

Function:

- Processes the equalization pulses during the vertical retract period to produce the V-Sync pulses.

Requirements:

- V-sync pulses should have reasonably sharp edges.
- Latency should be no more than 25% of one horizontal line period.

Inputs:

- Composite sync signal.

Outputs:

- Vertical sync pulse signal.

Horizontal and Vertical Sweep Generators

Function:

- Produce free-running horizontal and vertical ramp signals for raster scanning.

Requirements:

- Ramps should be sufficiently linear.
- Ramps must continue running when the NTSC input or sync pulses are absent.
- H- and V-ramp reset times should be short relative to their respective retrace periods.

Inputs:

- H-sync pulse signal for the horizontal sweep generator.
- V-sync pulse signal for the vertical sweep generator.

Outputs:

- $V_{H\text{-ramp}}$ to the oscilloscope X-axis.
- $V_{V\text{-ramp}}$ to the oscilloscope Y-axis.

Blanking Generator

Function:

- Produce a blanking flag during the horizontal and vertical retrace periods.
- Keep the flag active through the front porch, sync tip, and back porch.

Requirements:

- Blanking must cover the full horizontal and vertical retrace periods.
- Latency should be small relative to the horizontal retrace period.

Inputs:

- Horizontal sync signal.
- Vertical sync signal.

Outputs:

- Blanking flag signal for the luminance processing stage.

Luminance Separator

Function:

- Removes the synchronization information, leaving only the luminance information.

Requirements:

- Maintain sufficient bandwidth to preserve the luminance/image information.
- Minimize distortion of the luminance signal.

Inputs:

- Buffered and clamped NTSC composite video signal.

Outputs:

- Intermediate luminance signal with sync information removed.

Luminance Image Processing

Function:

- Combines the intermediate luminance signal with the blanking flag and allows adjustment of the brightness and contrast of the final image.

Requirements:

- Maximum and minimum output levels must match the oscilloscope Z-axis requirements.
- The black level during the blanking periods should be specified in terms of the chosen maximum/minimum levels of the luminance.
- Adjustment of the contrast and brightness should not change the black level.

Inputs:

- Intermediate luminance signal.
- Blanking flag.

Outputs:

- Final luminance signal V_{Lumin} for the oscilloscope Z-axis.

Design Activity

Power Supply Levels

We initially selected +5 V and +12 V supply channels to power the vertical/horizontal ramp generators (+12 V) and the remaining logic (+5 V). However, adding the luminance inverting amplifier required a negative supply, prompting a transition to a dual $\pm 12\text{V}$ supply rail. To maintain 5 V logic levels for the LM311 comparators, we derived +5 V from the +12 V rail using

a potentiometer voltage divider. Because direct current drawn from the other ICs(555 timer, LM311) caused loading effects and voltage fluctuations on this derived rail, we added an LM358 unity-gain buffer to isolate the divider and provide a stable, low-impedance +5 V supply.

Input Buffering, Clamping

A DC clamp is required for the NTSC signal to restore a fixed DC reference baseline due to the changing of the average voltage of the video waveform. Due to component constraints, specifically having only a single high-speed op-amp. We opted to use a passive clamp over an active clamp in order to save the high-speed op-amp to use for the amplifier in the luminance stage. Implementing an active clamp with a lower slew-rate op-amp would have severely degraded high-frequency video details and relying on a passive topology was the primary design trade-off we accepted to preserve overall signal integrity.

The clamps reference voltage is calculated by:

$$V_{ref} = 12 \cdot \frac{2.2k}{2.2k + 39k} = 0.641 V$$

Where V_D is was measured to be 0.62 V

$$V_{clamp} = V_{ref} - V_D = 0.641 - 0.62 = 0.021 V$$

Thus these resistor values for the voltage divider were chosen so the sync tips could be held slightly above 0V.

We choose to suppress the colour burst signal as we are not using chroma in this circuit and because we can get a more accurate reference voltage for the comparator in the composite sync separator. We are using a passive low pass filter rather than an active low pass filter due to component limitations mentioned earlier.

Since the NTSC color burst subcarrier is centered at 3.58 MHz we want to attenuate its amplitude without distorting critical baseband video components. A cutoff frequency of 1.59 MHz as the luminance signal is heavily concentrated on the lower end of the 4.2 MHz NTSC bandwidth.

$$|H(f)| = \frac{1}{\sqrt{1 + \left(\frac{f}{f_c}\right)^2}} = \frac{1}{\sqrt{1 + \left(\frac{3.58M}{1.59M}\right)^2}} = 0.4059$$

A cutoff frequency of 1.59 MHz gives us an amplitude reduction of the colour burst of around 59.4% ($1 - 0.4059$). This cutoff frequency effectively dampens the high-frequency colour burst while preserving fundamental luminance data.

Although having a video amplifier at the input stage is theoretically optimal for signal buffering and gain control, the raw NTSC input amplitude provided sufficient dynamic range for accurate measurement and sync extraction. Due to physical component constraints and to preserve subcircuit independence, input amplification was omitted in the physical implementation.

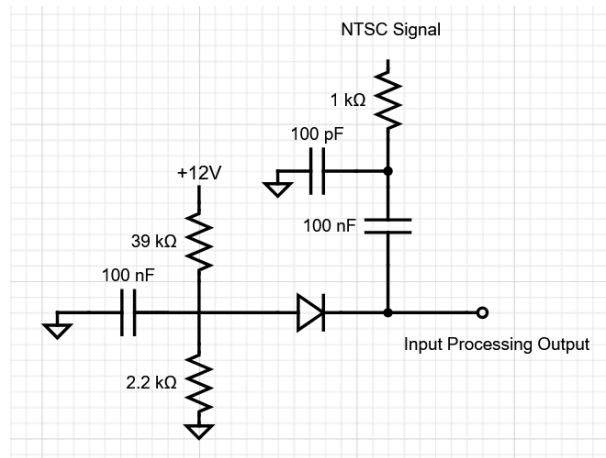


Figure 2: Input Clamping Circuit

Limitations of Passive Clamp with live NTSC Video

As shown in the oscilloscope capture, Channel 2 (cyan) displays the clamped waveform using the diode clamp circuit, while Channel 1 (yellow) represents the synthesized NTSC input signal. Although the passive clamp operates effectively with the controlled artificial NTSC signal, testing with a live NTSC source revealed stability issues. Due to rapid variations in Average Picture Level and dynamic video content in real-world signals, the passive clamp's time constant prevents it from maintaining a constant DC baseline, causing the clamped level to drift or swing.

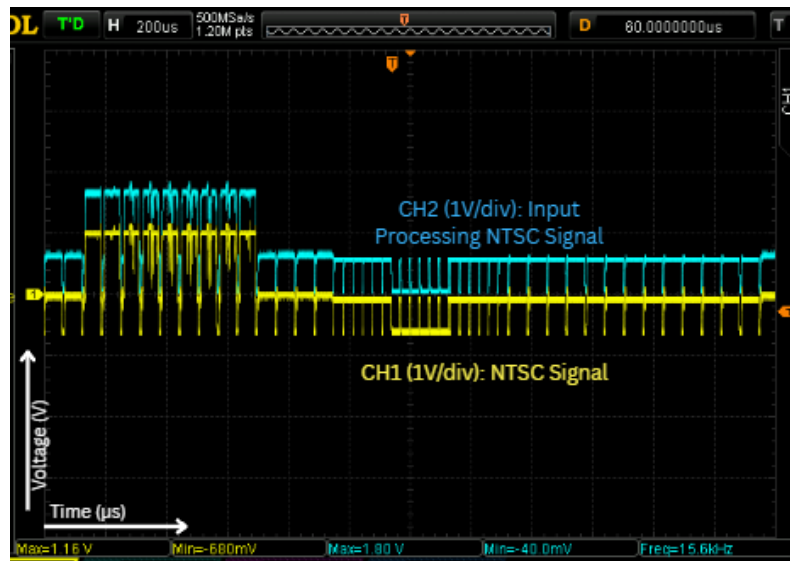


Figure 3: NTSC Passive Clamp Capture

Passive Clamp Loading Anomaly and System Integration

When integrating the passive clamp with the subsequent composite sync (C-sync) separator stage, an unexpected loading anomaly occurred: the C-sync separator failed to operate unless an oscilloscope probe was actively connected to the clamp's output node. This behavior suggested that the clamp output was presenting an unstable, high-impedance node to the comparator, lacking an adequate DC discharge path and allowing parasitic noise to disrupt proper triggering. The 10M Ω input resistance and parallel capacitance of the oscilloscope probe provided the necessary bleed path and high-frequency filtering required to stabilize the node. To resolve this issue in the hardware, a 1M Ω resistor was added from the clamp output to ground, providing a permanent discharge path that restored proper C-sync separator operation without requiring external probing.

Composite Sync Separator

First Iteration

For our composite sync separator design we opted to use a LM311 comparator due to its low latency and single channel supply. Since the LM311 comparator has an open collector output we need a pull up resistor. We decided to use an active low polarity as we want the output to remain HIGH by default and drop LOW upon detecting a sync pulse. Thus the inverting terminal will be set as our reference voltage and the non-inverting terminal will be the input from the Input Buffering, Clamping stage.

From measuring from the sync tip (4 mV) to right below the colour burst (348 mV)

$$V_{ref} = \frac{348 + 4}{2} = 176 \text{ mV}$$

Calculating the voltage divider

$$176 \text{ mV} = 5 \cdot \frac{R_2}{R_2 + R_1}$$

$$R_1 = 27.41 R_2$$

Choosing $R_2 = 1 \text{ k}\Omega$, $R_1 = 27 \text{ k}\Omega$

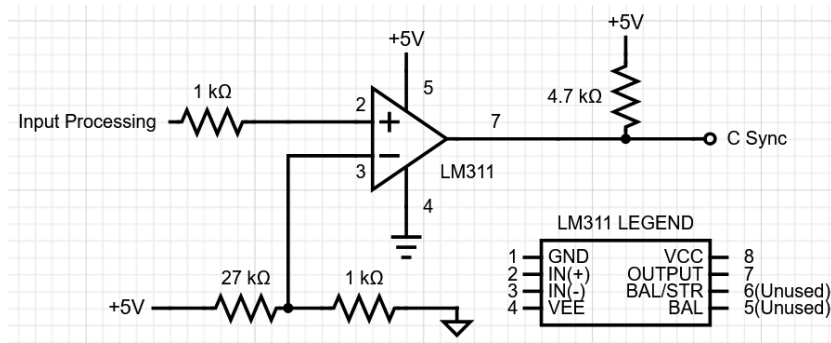


Figure 4: First Iteration C-Sync Separator Circuit

Second Iteration

Although the first iteration subcircuit proved sufficient with only the input buffering/clamping stage connected, when connected to the h-sync separator subcircuit the output signal of the c-sync separator became unstable. We realized after looking at the data sheet for the LM311 the resistor used for the pull up resistor was too high, so we swapped to a 220Ω resistor instead. We also added a positive feedback resistor (1MΩ) in order to avoid small noise spikes that would cause the comparator output to rapidly flip on and off. After these changes our output became stable again.

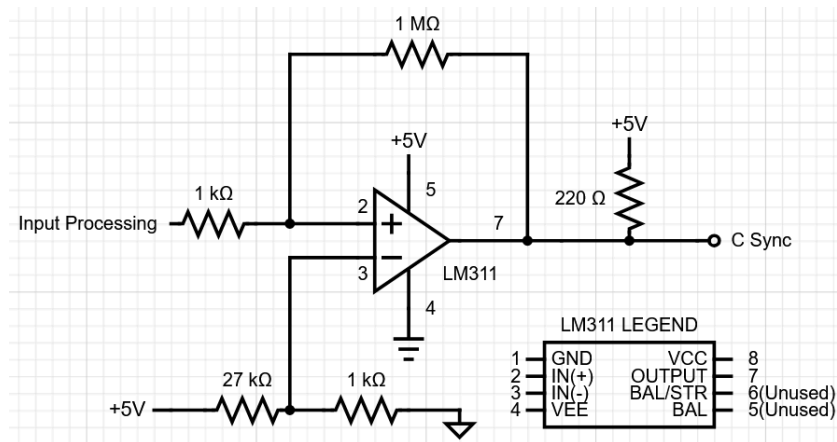


Figure 5: Second Iteration C-Sync Separator Circuit

Vertical Sync Separator

First iteration

Since the vertical sync pulse has a frequency of 59.94 Hz we decided to use a low pass filter with a cutoff frequency of 1591 Hz to isolate the vertical sync pulses. We decided a passive low-pass filter was preferable by prioritizing circuit simplicity and stability over active gain. By using only a resistor and capacitor, it eliminates additional op-amps and active feedback stages, reducing board space and power consumption.

$$1591 = \frac{1}{2\pi RC}$$

$$RC = \frac{1}{9996.54}$$

Choosing $R = 10\text{k}\Omega$, $C = 10\text{nF}$

Although the passive filter yields gentler roll-off slopes, this is negligible in this application because the integrated output is directly fed into a second stage utilizing a 555 timer with the inputs pins THRESH and TRIG connected to create hysteresis to prevent chattering and restore sharp digital transitions. A 10 nF decoupling capacitor was added to pin 5 to ground to filter out noise.

Second Iteration

Although a 1,591 Hz cutoff frequency is theoretically ideal for isolating vertical sync pulses, testing revealed that the heavily attenuated output spikes were insufficient to trigger the 555 timer. Replacing the 10 nF capacitor with a 1 nF capacitor increased the cutoff frequency to 15.9 kHz, significantly boosting the pulse amplitude for reliable 555 triggering. While this higher cutoff frequency allowed 15.734 kHz horizontal serration ripples to pass through, these

high-frequency dips were effectively rejected by the secondary 555 stage due to its built-in hysteresis thresholds, preserving clean vertical sync generation.

Horizontal Sync Separator

First Iteration

For the horizontal sync separator since the horizontal sync pulse has a frequency of 15.734 kHz our first design used a similar approach to the vertical sync separator but instead using a passive high pass filter with a cutoff frequency of 15.915 kHz as the first stage then into a second stage using a 555 timer with hysteresis to produce the h-sync separator output.

$$15915 = \frac{1}{2\pi RC}$$

$$RC = \frac{1}{99996}$$

Choosing $R = 1\text{k}\Omega$, $C = 10\text{nF}$

Although this method worked for the vertical sync separator, it ultimately failed for the horizontal sync separator because we only want to capture the 15.734 kHz frequency of the h-sync pulse but during the vertical blanking interval the NTSC introduces equalizing pulse and vertical serration at twice the horizontal sync pulse frequency 31.468 kHz.

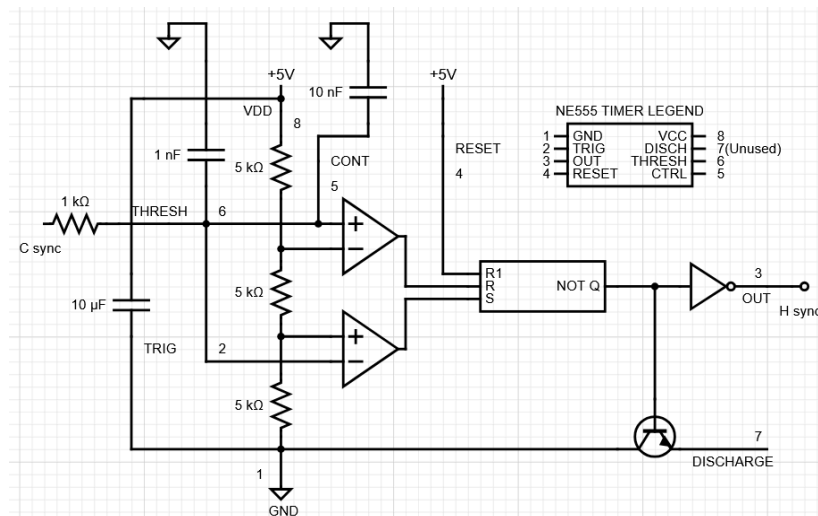


Figure 6: First Iteration H-Sync Circuit

Second Iteration

For our second approach we decided to use a monostable multivibrator constructed from 555 timers. To configure a 555 timer as a monostable multivibrator, we need to connect the DISCHARGE pin and THRESHOLD pin together to a node between a capacitor and resistor to

charge the capacitor. Therefore we must use the RC charging formula to determine the width of the pulse.

$$V(t) = V_{CC} - (V_{CC} - V_{initial})e^{-\frac{t}{RC}}$$

Since the DISCHARGE and THRESHOLD pins are connected we want to calculate the charge time from 0V to $\frac{2}{3} V_{CC}$. Therefore, $V(t) = \frac{2}{3} V_{CC}$, $V_{initial} = 0$.

$$\frac{2}{3}V_{CC} = V_{CC} - V_{CC}e^{-\frac{t}{RC}}$$

Solving for t gives us

$$t = \ln(3) \cdot RC = 1.0986 \cdot RC$$

By setting its output pulse duration to approximately 45 μ s, a monostable multivibrator creates a blind window that naturally filters out double-frequency pulses while passing normal horizontal sync. Standard horizontal lines arrive every 63.5 μ s (15.7 kHz), so when a line edge triggers the timer at $t = 0$ μ s, the output stays active for 45 μ s, finishes, and resets in time to catch the next line pulse at 63.5 μ s. However, during vertical blanking, equalizing and serration pulses arrive at twice the rate at every 31.75 μ s (31.5 kHz). When the first edge fires the timer, the double-frequency pulse arrives at 31.75 μ s while the monostable is still in its active 45 μ s period, causing the circuit to completely ignore the extra edge. Because 45 μ s sits cleanly between 31.75 μ s and 63.5 μ s, the timer masks out every intermediate pulse and preserves a continuous 15.7 kHz output rate across the entire frame.

$$\tau = 1.1(3.9k)(10n) = 42.9 \mu s$$

Choosing $R = 3.9k\Omega$, $C = 10nF$

But this design produced an unstable output and unwanted feedback into the previous subcircuit stage, introducing unwanted noise. To resolve this instability, we used an LM311 comparator stage between the composite sync output and the H-sync separator input to serve as a digital buffer. Using an equal-resistor voltage divider, we set the LM311 reference voltage to 2.5 V, the midpoint of our 0 V to 5 V C-sync signal, ensuring noise rejection and stage isolation. Although this resolved the feedback problem, it did not fix the unstable output

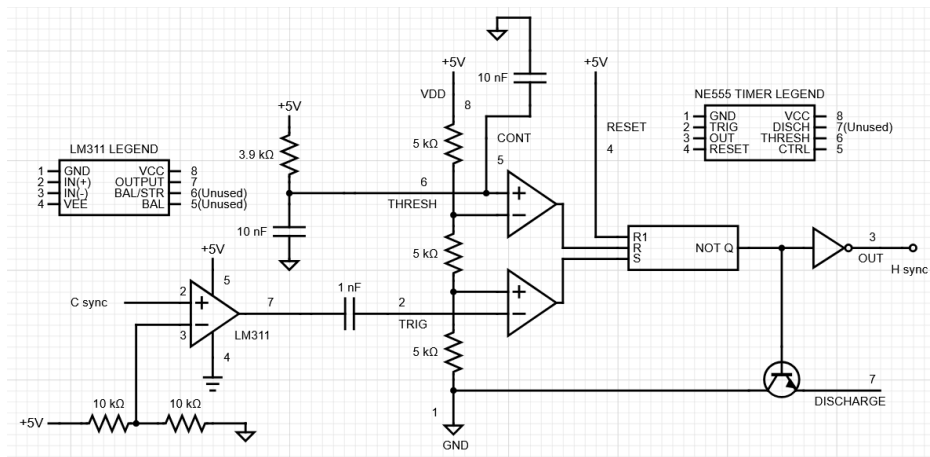


Figure 7: Second Iteration H-Sync Circuit

Third Iteration

In this third design we attempt to fix the instability issue by using a dual-stage masking approach using monostable multivibrators. Dual-stage masking fixes the instability issue by dividing the workload between two interlocked timers: one dedicated exclusively to establishing a time-masking window and the other to generating the clean output pulse. In a single 555 timer monostable, attempting to mask out double-frequency pulses while simultaneously triggering off raw, level-sensitive sync edges causes trigger lockup and output jitter. The dual-stage setup solves this because Timer 1 fires first to create a 42.9 μs masking window that activates an external NPN transistor, which physically locks out the second timer and creates a dead zone during which any 31.468 kHz double-frequency pulses at 31.75 μs are completely blocked. By actively clamping Timer 2 during this window, the circuit prevents overlapping trigger conflicts, allows the timing capacitors to fully discharge back to 0 V, and ensures Timer 2 only fires on valid 15.7 kHz line edges, eliminating output chattering, jitter, and pulse-width drift across the vertical blanking interval.

Timer 1 pulse width:

$$t = 1.1(39k)(1n) = 42.9 \mu\text{s}$$

Timer 2 pulse width:

$$t = 1.1(47k)(1n) = 51.7 \mu\text{s}$$

To ensure reliable functionality an AC-coupling network consisting of a 1 nF series capacitor, a 10 kΩ pull-up resistor, and a clamping diode was implemented at each trigger pin to prevent trigger lockup by turning raw sync edges into short spikes, while clamping positive voltage overshoot to protect the chip from ringing and false re-triggering.

While our application relies strictly on the falling edges of the pulses for line timing, the trade-off of generating a 51.7 μs output pulse is that it consumes nearly majority of the 63.55 μs horizontal line period, drastically shortening the timing capacitor's recovery window to just 11.8 μs and stripping away the standard video line structure. Furthermore, because this extended pulse completely eliminates the front and back porches, it breaks compatibility with circuits that depend on a standard 4.7 μs pulse width and a dedicated back-porch sampling interval.

The full third iteration H-sync circuit diagram can be found in the appendix Figure 35.

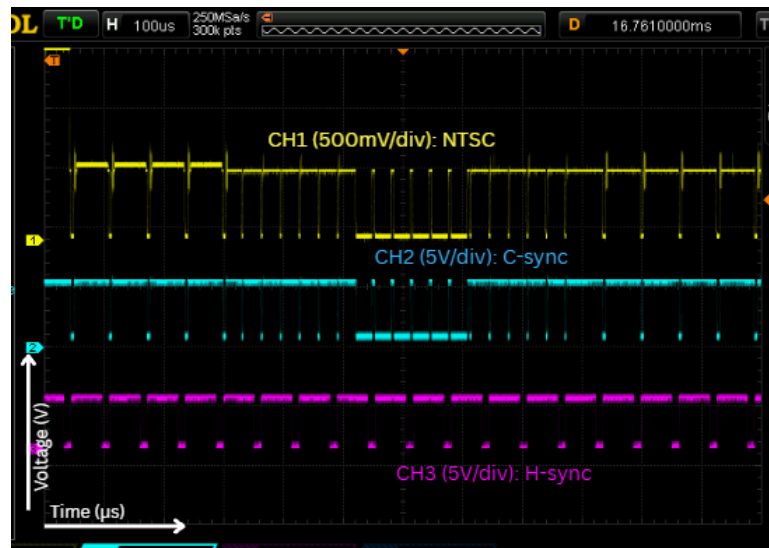


Figure 8: C-sync and H-sync Oscilloscope Captures

Fourth Iteration

As shown in the third iteration, the generated pulse width measured 51.7 μs due to a component selection error, as a 47 k Ω resistor was mistakenly installed instead of the intended 4.7 k Ω value. Correcting the resistor value reduces the RC time constant by a factor of ten, yielding the expected 5.17 μs pulse width. You can see the changes in the screen capture below.

Timer 2 pulse width:

$$t = 1.1(4.7k)(1n) = 5.17 \mu\text{s}$$

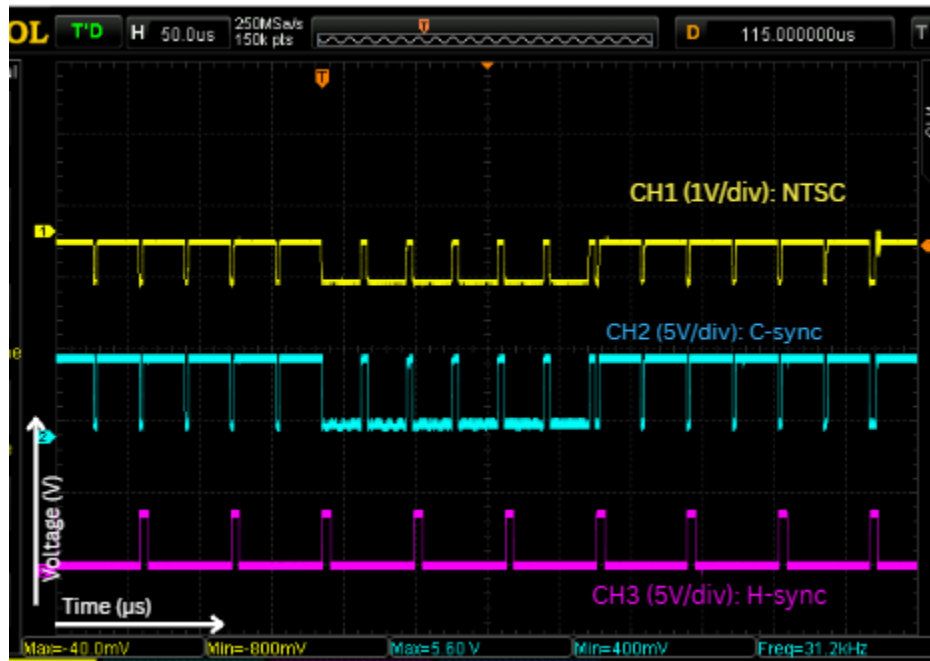


Figure 9: Fourth Iteration H-sync Oscilloscope Capture

Horizontal and Vertical Sweep Generator

First Iteration

For our horizontal and vertical sweep generator we used 555 timers as we can take advantage of the internal discharge switch and threshold comparators to create our ramps. We can create our ramp generators by measuring the voltage of a capacitor as current from a resistor charge it and by connect them to the DISCHARGE pin we can allow the voltage in the capacitor to discharge to ground through this pin and by connecting the DISCHARGE pin to the TRIG pin this allows the capacitor to recharge whenever the capacitor dissipates to $\frac{1}{3} V_{cc}$. But since we are using a resistor to charge up the capacitor, the voltage in the capacitor does not charge up linearly. So we will use a smaller section of the voltage curve so it will seem more linear. Thus we connect the output of the V-sync or H-sync separators into the THRESHOLD pin of the 555 timer. We are going to use an AC coupling capacitor at the output using a 100 nF capacitor and at the input using a 10 nF at pin 5.

Since the vertical sync pulses have a frequency of 59.94 Hz, we want the capacitor to charge to 1V above $\frac{1}{3} V_{cc}$ in 16.7 ms ($1/59.94$). To calculate the component values we first take the RC charging formula.

$$V(t) = V_{cc} - (V_{cc} - V_{initial})e^{-\frac{t}{RC}}$$

And if we want to start charging from $\frac{1}{3} V_{CC}$ to $1 + \frac{1}{3} V_{CC}$. Then $V_{initial} = \frac{1}{3} V_{CC}$, $V(t) = 1 + \frac{1}{3} V_{CC}$
Substituting the desired voltage ranges.

$$1 + \frac{V_{CC}}{3} = V_{CC} - (V_{CC} - \frac{V_{CC}}{3})e^{-\frac{t}{RC}}$$

Solving for t give us

$$t = RC \cdot \ln\left(\frac{2V_{CC}}{2V_{CC} - 3}\right)$$

Since we want $t = 16.7 \text{ ms}$, $V_{CC} = 5\text{V}$ choose $C = 10\mu\text{F}$

$$R = \frac{\tau}{C \cdot \ln\left(\frac{2 \cdot V_{CC}}{2 \cdot V_{CC} - 3}\right)} = \frac{16.7m}{(10\mu) \ln\left(\frac{10}{10-3}\right)} = 4682 \Omega$$

Thus choose $R = 4.7k\Omega$

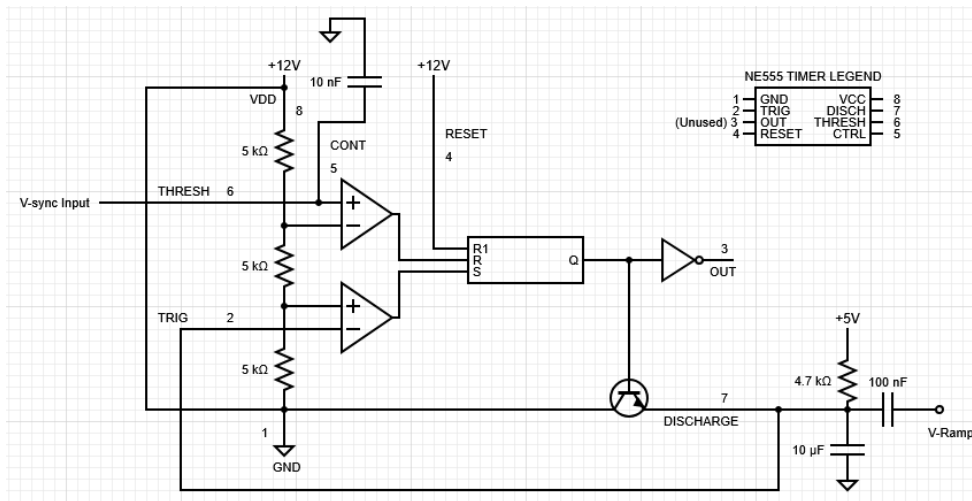


Figure 10: First Iteration H-Ramp/V-Ramp Circuit

Second Iteration

Although the first iteration subcircuit provided sufficient ramps with slight but noticeable curvature in the slope, we can greatly improve the linearity of the slope just by replacing the resistor with a wilson current mirror and if we increase the voltage amplitude of the ramps the noise relative to the ramp will look smaller. Since we are increasing the supply voltage the threshold level will also increase, thus we require a DC offset at the input to pull the voltage up. Since we decided to choose to increase the supply voltage to 12V, the threshold level increased to $8V(\frac{2}{3} V_{CC})$. We will use a voltage divider to bring up the idle voltage.

$$V = 12 \cdot \frac{15k}{15k+10k} = 7.2V$$

Here we choose to use a 15kΩ and 10kΩ resistor to bring up the idle voltage to 7.2 V, right below the threshold level. We choose to charge the capacitor up to 6V, recalculating the capacitor and resistor value using the formula for charging a capacitor with a constant current source.

$$t = \frac{C \cdot \Delta V}{I}$$

Using $t = 16.7\text{ms}$, $\Delta V = 6$, choosing $C = 10\mu\text{F}$

$$I = \frac{C \cdot \Delta V}{t} = \frac{(10\mu)(6)}{16.7m} = 3.59mA$$

Calculating resistance needed for wilson current mirror

$$R = \frac{12 - 0.7 - 0.7}{3.59m} = 2952\Omega$$

Thus choose $R = 3.3k\Omega$

Since H sync pulses have a frequency of 15.734 kHz, we want the capacitor to charge up 6V in $t = 63.5\mu\text{s}$ ($1/15.734k$). Choosing $C = 47\text{nF}$.

$$I = \frac{C \cdot \Delta V}{t} = \frac{(47n)(6)}{63.5\mu} = 4.44mA$$

Calculating resistance needed for the Wilson current mirror:

$$R = \frac{12 - 0.7 - 0.7}{4.44m} = 2387\Omega$$

Thus choose $R = 2.2k\Omega$.

The full second iteration H-ramp and V-ramp circuit diagram can be found in the appendix Figure 36.

H-sync Separator Fix

Although no direct modifications were made to the horizontal ramp generator, resolving the issues within the horizontal sync separator restored proper ramp resetting. Fixing the horizontal sync separator ensured that reliable reset pulses reached the ramp generator, eliminating the failure where the ramp would not discharge, as demonstrated in the before and after waveforms below.

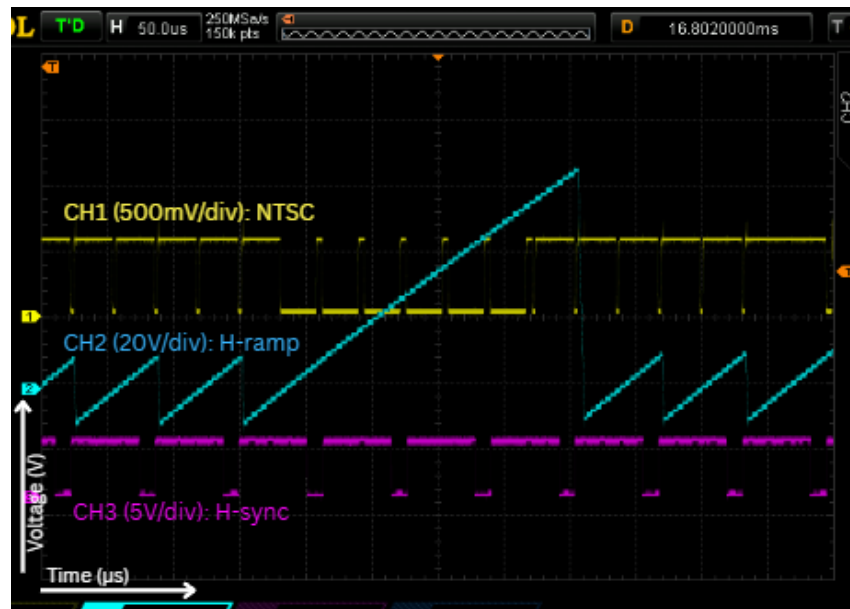


Figure 11: Before H-sync Separator Change

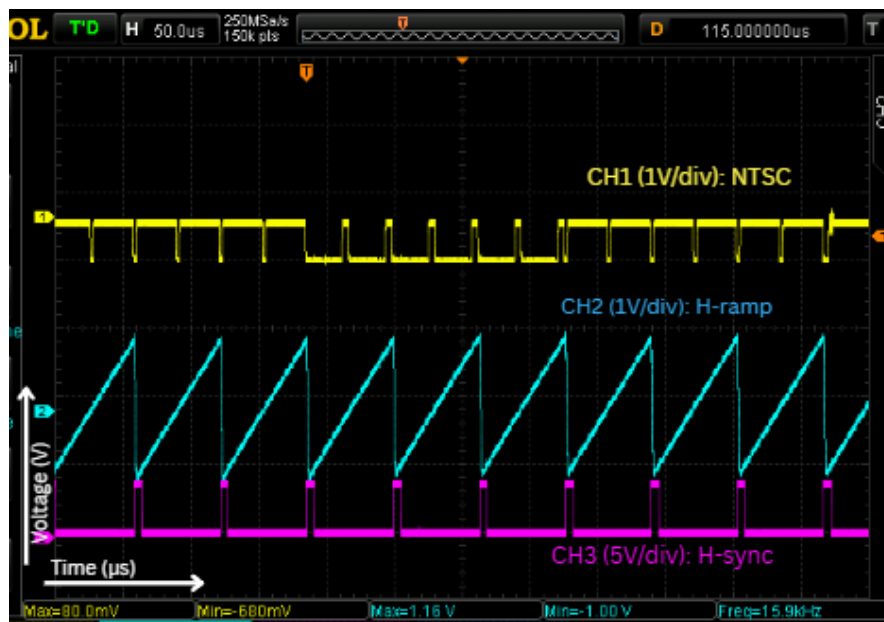


Figure 12: After H-sync Separator Change

Blanking Generator

Although a functional blanking generator was not successfully implemented in the final circuit. However, the intended design would generate a digital blanking flag that becomes active during both the horizontal and vertical retrace intervals. o the luminance-processing path. According to

the project specification, this flag must remain active throughout the front porch, synchronization pulse, and back porch, rather than only during the synchronization pulse itself.

The horizontal and vertical sync signals could each trigger a monostable timing circuit. The horizontal monostable would generate a pulse wide enough to cover the complete horizontal retrace interval, while the vertical monostable would generate a longer pulse covering the complete vertical retrace interval. These two signals would then be combined using an OR function.

$$B_{blank} = B_H + B_V$$

B_H represents the horizontal blanking pulse and B_V represents the vertical blanking pulse. This would be so that blanking is active whenever either horizontal or vertical retrace is occurring. The resulting blanking flag would be sent to the luminance image-processing stage, where it would override the luminance signal and force the Z-axis output to the chosen black level. This is consistent with the specified system architecture, where the blanking generator feeds directly into

Luminance Separator

Since this subcircuit requires the removal of the synchronization information while only leaving the luminance information, we choose to use a clipper circuit because sync pulses extend below the black level.

With a V_D of 0.62 V, we can calculate the clipping level as

$$V_{ref} = 12 \cdot \frac{2.2k}{39k+2.2k} = 0.641 \text{ V}$$

$$V_{clip} = V_{ref} - V_D = 0.641 - 0.62 = 0.02 \text{ V}$$

Thus the clipping level is 0.02 V

Since the majority of the luminance frequencies are on the lower end of the 4.2 MHz NTSC bandwidth, we decided to choose a bandwidth of 0 to 1.59MHz. To do this we use a passive low pass filter with a cutoff frequency of 1.59 MHz in order to isolate the luminance part from the chroma

$$1.59 \text{ MHz} = \frac{1}{2\pi RC}$$

Choosing $R = 100\Omega$

$$C = \frac{1}{2\pi(100)(1.59M)} = 1nF.$$

While selecting a 1.59 MHz cutoff frequency effectively attenuates the 3.58 MHz chroma color subcarrier, it restricts the luminance bandwidth well below its standard 4.2 MHz limit. This trade-off was intentionally made to prioritize stable sync separation and a subcarrier-free video baseline over high-frequency spatial details, resulting in a slightly softened image on the display.

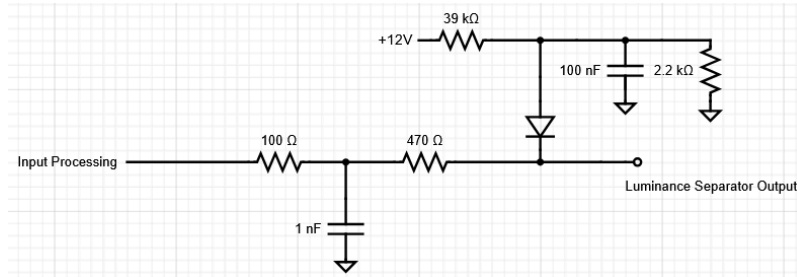


Figure 13: Luminance Separator Circuit

Luminance Image Processing

The luminance image processing stage should be able to adjust the contrast and brightness of the video with the use of trimpots, where contrast represents the gain and brightness represents the DC offset. Since we are driving a CRT cathode, we also need to invert the luminance signal because the CRT tube operates with an inverted brightness characteristic.

We used the LM6171 for the inverting amplifier due to its high bandwidth and slewrate, for the resistor we chose to use a 10k ohm pot in series with a 1k ohm resistor for the inverting terminal and a 100k ohm feedback resistor. These resistor values give us a gain range of - 9.1 V/V to -100 V/V which is sufficient enough for a wide range of contrast adjustment.

$$A_{max} = - \frac{R_f}{R_1} = - \frac{100k}{1k} = - 100 V/V$$

$$A_{min} = - \frac{R_f}{R_1} = - \frac{100k}{10k+1k} = - 9.1 V/V$$

While the brightness can be controlled by adjusting the DC offset of the output of the amplifier. To achieve this we connect the output to the wiper of the trim pot and connect the other 2 nodes to 12V and ground through a 10k resistor. This creates a DC offset range of 6V to 12V.

$$V_{OS,min} = 12 \cdot \frac{10k}{10k} = 12 V$$

$$V_{OS,min} = 12 \cdot \frac{10k}{10k+10k} = 6V$$

The output V_{Lumin} can be characterized by the equation.

$$V_{Lumin} = A * V_{Lumin Sep} + V_{OS}$$

Where A is the gain from the amplifier, $V_{Lumin Sep}$ is the incoming signal from the luminance separator, and V_{OS} is the offset.

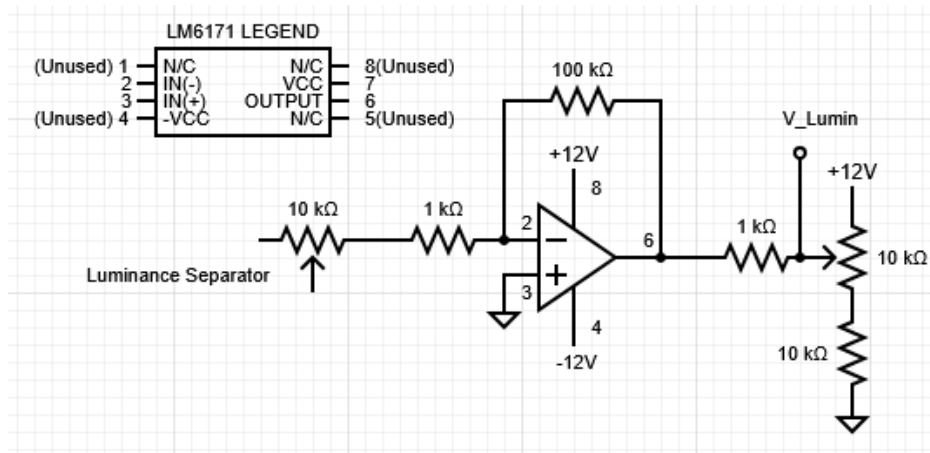


Figure 14: Luminance Image Processing Circuit

The Final System

Input Buffering, Clamping

Theory of Operation

The input buffering, and clamping stage works by first filtering the incoming video signal using a passive low pass filter with a cutoff frequency at 1.59 MHz attenuating the colour burst signal but also preserving the luminance signal. Then we use a diode clamper with a clamp voltage of 0.021V so when the incoming NTSC falls below 0.021V the diode turns on pulling up the signal to 0.021V.

Measurement Procedure

1. Apply the artificial NTSC signal to the input clamping circuit and observe both the original input and processed output on the oscilloscope.

2. Measure the minimum voltage of the processed waveform and verify that the sync tips are clamped near the designed value of 0.021 V.
3. Compare the input and output waveforms to confirm that the 1.59 MHz low-pass filter attenuates the colour-burst/high-frequency components while preserving the main luminance waveform.
4. Verify that the processed artificial NTSC signal remains stable and sufficiently undistorted for use by the following composite sync separator and luminance stages.

Measurement Results

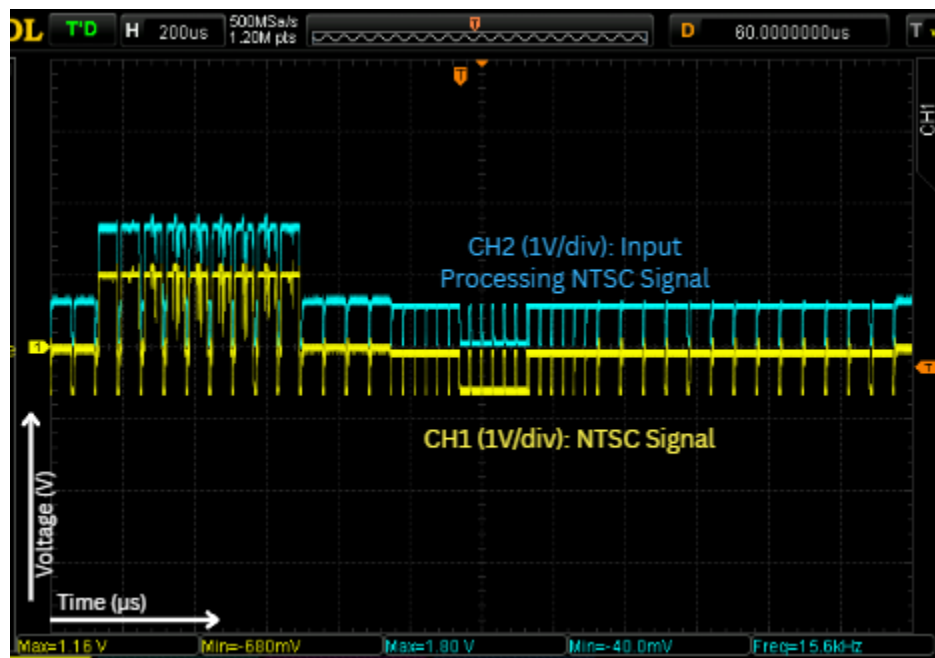


Figure 15: Input Processing NTSC and Artificial NTSC Oscilloscope Capture

Interpretation of Results

Disclaimer: This analysis applies exclusively to the circuit's performance with the synthetic NTSC signal; as noted previously in the design activity section for this subcircuit, the passive clamp experiences DC baseline drift when processing a live, dynamic NTSC video feed.

With both channels zero-positioned to the same baseline reference on the scope, Channel 2 (cyan) clearly shows the incoming NTSC video signal (Channel 1, yellow) shifted upward so that its lowest sync tips are clamped near 0V (ground). The clamped waveform perfectly preserves all structural components of the original signal, including active video, horizontal sync pulses, and the vertical sync interval, while successfully establishing the positive DC reference needed for downstream comparator processing.

Composite Sync Separator

Theory of Operation

The composite sync separator works by slicing the incoming video signal at a precise threshold to isolate the synchronization pulses from the active video content. With the reference voltage set to 176 mV at the inverting terminal (-), the comparator operates in an active-low configuration. The output is normally pulled up to 5 V, but whenever the non-inverting input (+) exceeds 176 mV, the comparator drives the output to 0 V (ground).

Measurement Procedure

1. Apply the NTSC signal and display the NTSC waveform and composite sync output on separate oscilloscope channels.
2. Measure the LM311 reference voltage and verify that the switching level is positioned between the sync-tip level and the blanking level of the clamped video signal.
3. Compare the two waveforms and confirm that each synchronization portion of the NTSC signal produces a corresponding composite-sync pulse, including the horizontal, equalization, and vertical sync pulses.
4. Vary the video content and observe the composite-sync output to ensure that the image information and colour burst do not produce unwanted pulses or missed sync transitions.

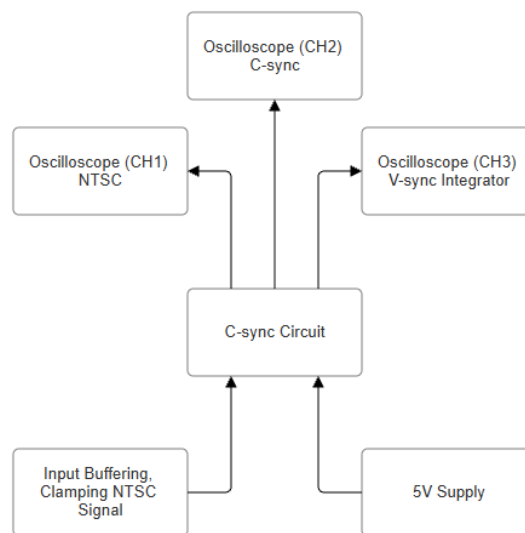


Figure 16: C-sync Block Diagram Connections

Measurement Results

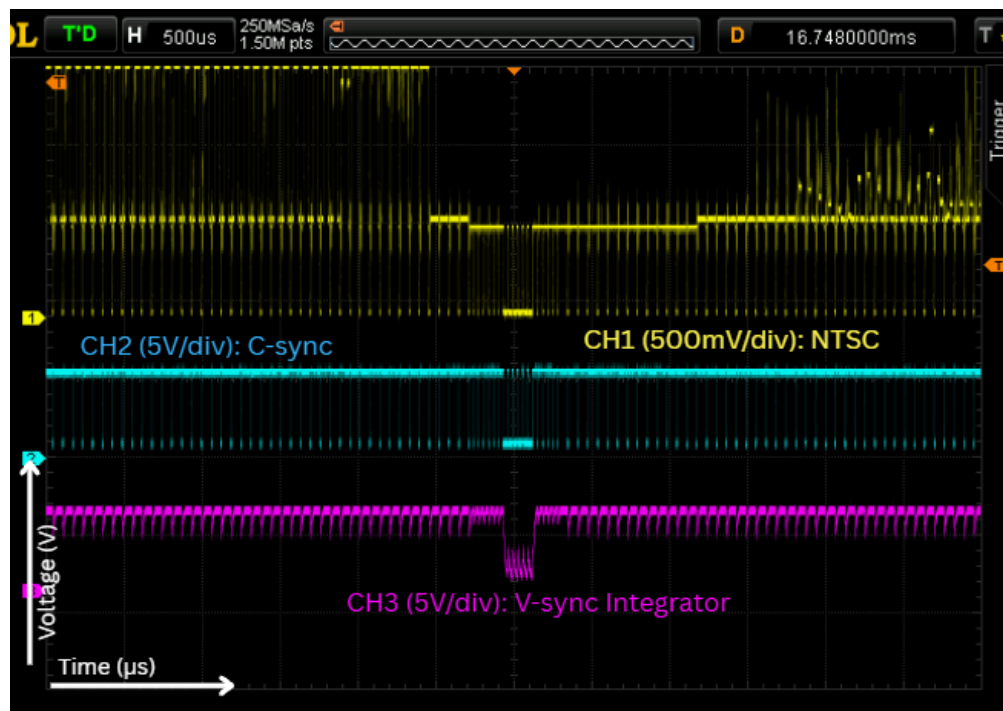


Figure 17: C-sync and V-sync Integrator Oscilloscope Capture

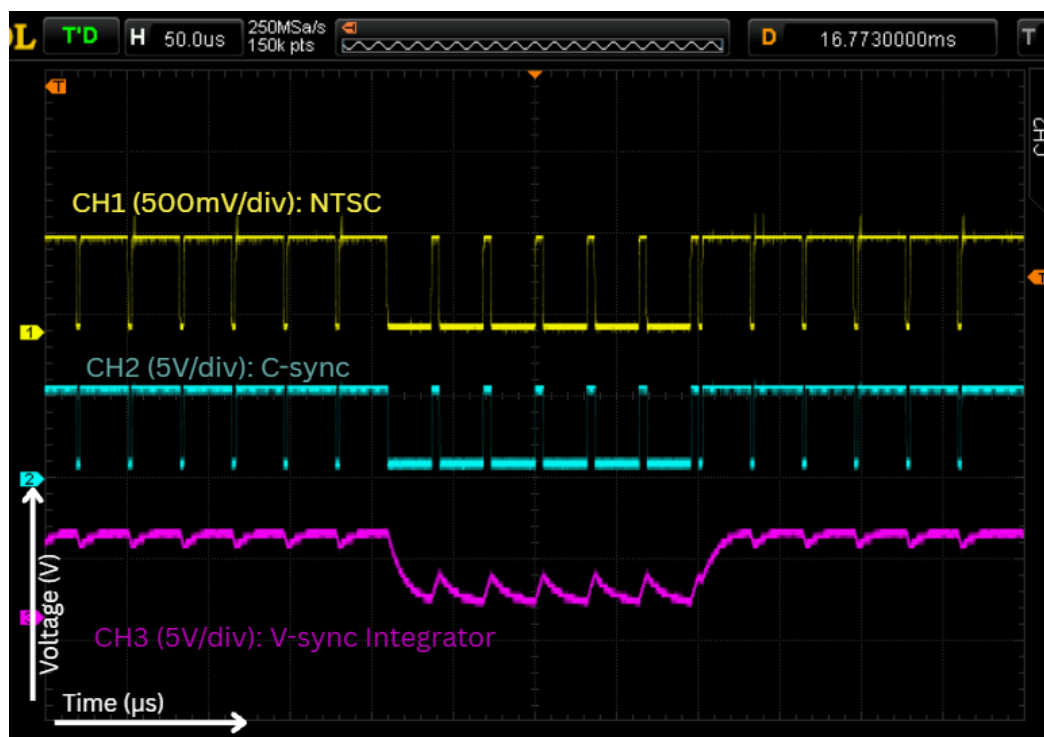


Figure 18: Zoomed C-sync and V-sync Integrator Oscilloscope Capture

Interpretation of Results

The composite sync separator demonstrates highly reliable pulse extraction, producing clean, sharp 5V logic transitions that reliably isolate synchronization pulses from the incoming NTSC video signal. The extracted output (Channel 2) successfully strips away all active luminance content while maintaining precise timing alignment with the input sync tips (Channel 1) during both horizontal scanning lines and the vertical retrace interval. Furthermore, the sharp pulse edges and well-defined high/low voltage levels show minimal jitter or threshold chatter, proving that the comparator stage processes equalization and vertical sync pulses accurately without false triggering or pulse degradation.

Horizontal Sync Separator

Theory of Operation

The first monostable multivibrator generates a 42.9 μs pulse to mask the double-frequency equalization pulses (31.75 μs period) during the vertical blanking interval, while leaving the standard 63.5 μs horizontal sync period undisturbed. The second monostable's TRIG pin (Node 8) is connected to both C-sync and the emitter of an NPN transistor, creating a hardware AND logic condition for triggering. The first condition requires the output of the first monostable to be LOW; when its output is HIGH, it turns on the NPN transistor (configured as an emitter follower with its collector at +5V), forcing Node 8 up to approximately 4.3V and overriding any incoming trigger edges. The second condition requires a falling edge from C-sync to pull Node 8 below $\frac{1}{3}V_{CC}$ (1.67V). When both conditions are met, Timer 1 is LOW and a C-sync falling edge arrives, the second monostable triggers, generating a pulse width of 5.17 μs .

Measurement Procedure

1. Probe C-Sync and the output first monostable of H-Sync to see that all the double frequencies have been masked.
2. Probe the C-sync signal alongside the outputs of the first and second H-sync monostable stages to verify that the second monostable fires exclusively when the first monostable output is LOW and during a horizontal sync tip.
3. Confirm the H-sync frequency is 15.734 kHz by measuring the period during a normal line.
4. Use the oscilloscope to determine the H-Sync pulse duration and verify that it is close to 4 μs .

5. Observe the signal during the vertical retrace period and check that the H-Sync pulses remain continuous without any skipped or additional pulses through the equalization and serration intervals.
6. Compare the timing of the Composite Sync transition with the corresponding H-Sync transition and measure the delay between them. Verify that this delay is negligible compared with the duration of one horizontal line.

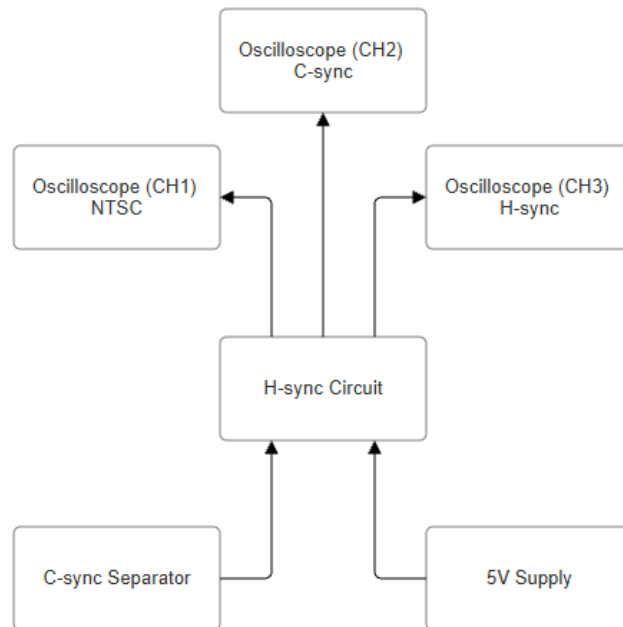


Figure 19: H-sync Block Diagram Connections

Measurement Results

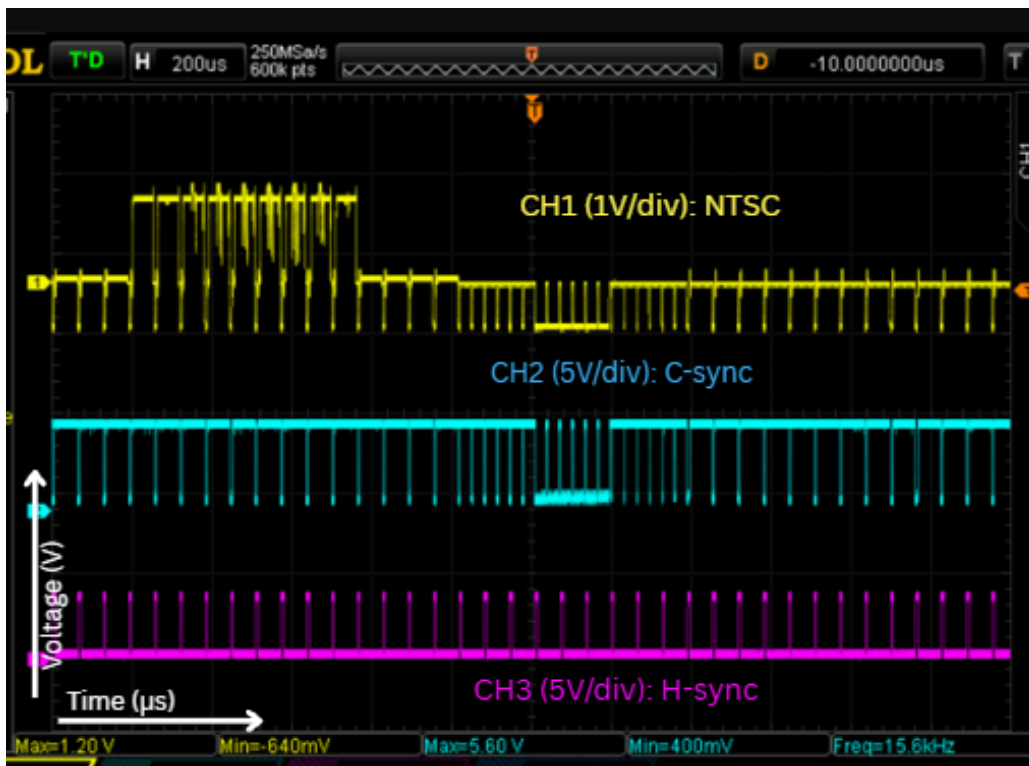


Figure 20: H-sync and C-sync Oscilloscope Capture

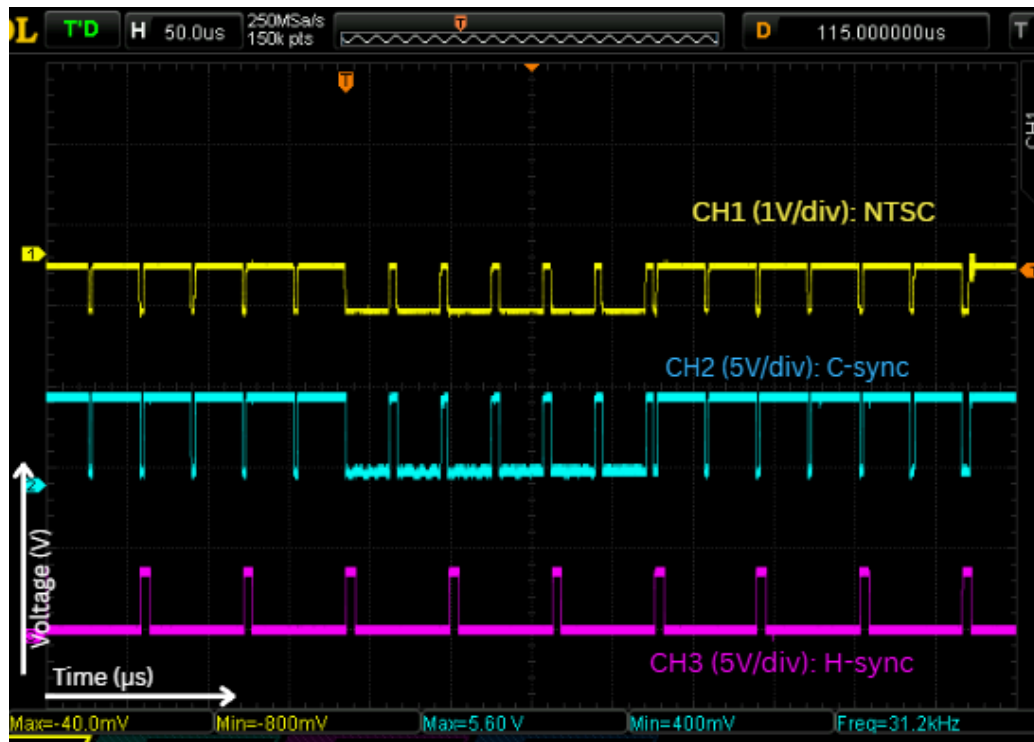


Figure 21: Zoomed H-sync and C-sync Oscilloscope Capture

Interpretation of Results

The output on Channel 3 (magenta) successfully isolates a continuous, uniform stream of horizontal sync pulses from the composite signal, maintaining a consistent frequency (15.734 kHz) even through the vertical retrace interval. A slight propagation delay is visible between the falling edge of the input composite sync signal (Channel 2, cyan) and the rising edge of the separated horizontal sync pulse (Channel 3, magenta), this delay is acceptable as it remains well within the timing constraints of the overall system design.

Vertical Sync Separator

Theory of Operation

The V-sync separator works by filtering out high-frequency horizontal sync pulses and extracting only the longer vertical synchronization interval. The C-sync signal passes through a low-pass filter with a cutoff frequency of 1591 Hz, attenuation that allows only the broader V-sync pulse energy to remain. The filtered output is then fed into a 555 timer stage to add hysteresis, preventing output chattering or false triggers caused by residual high-frequency noise.

Measurement Procedure

1. Display the composite sync signal and the V-sync output from the NE555 on separate oscilloscope channels over several video fields.

2. Measure the repetition rate of the V-sync output and verify that it occurs at approximately 60 Hz.
3. Check that the circuit produces a single V-sync pulse for each field and does not respond to the normal horizontal sync pulses.
4. Measure the delay between the vertical-sync portion of the composite sync waveform and the generated V-sync pulse, and verify that the delay remains within the specified 25% of one line period.
5. Change the video content and confirm that the vertical sync output remains stable without extra or missing pulses.

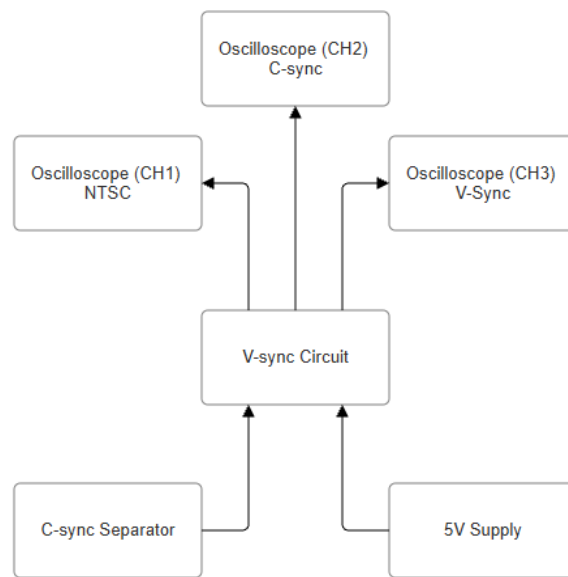


Figure 22: V-sync Block Diagram Connections

Measurement Results

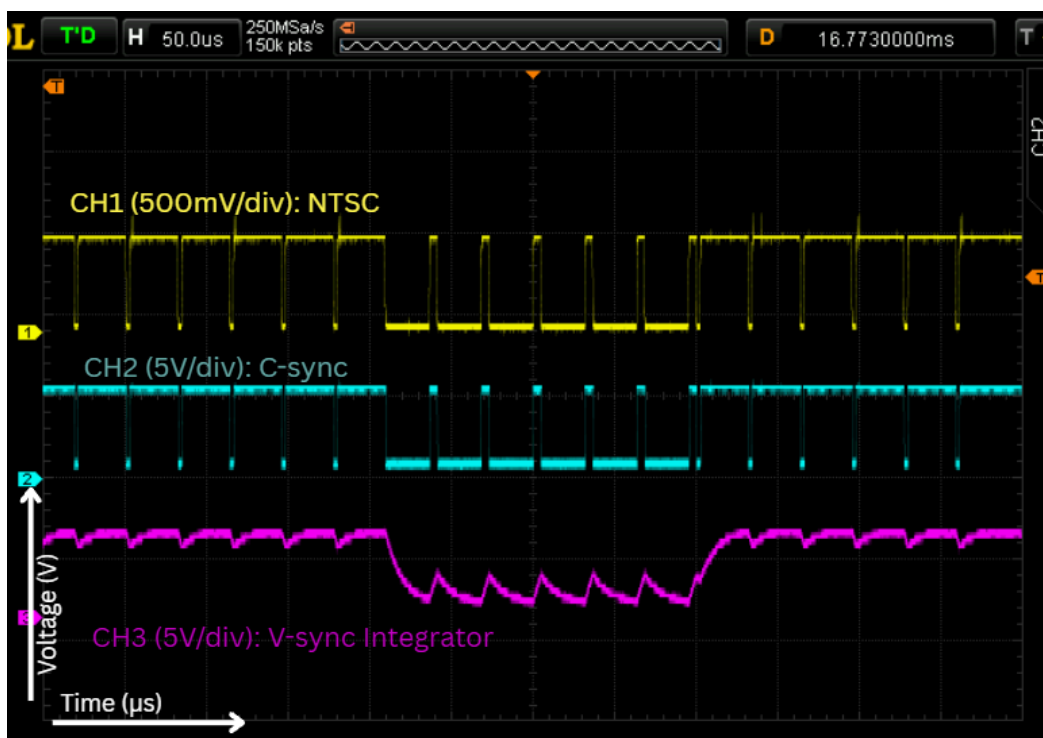


Figure 23: C-sync and V-sync Integrator Oscilloscope Captures

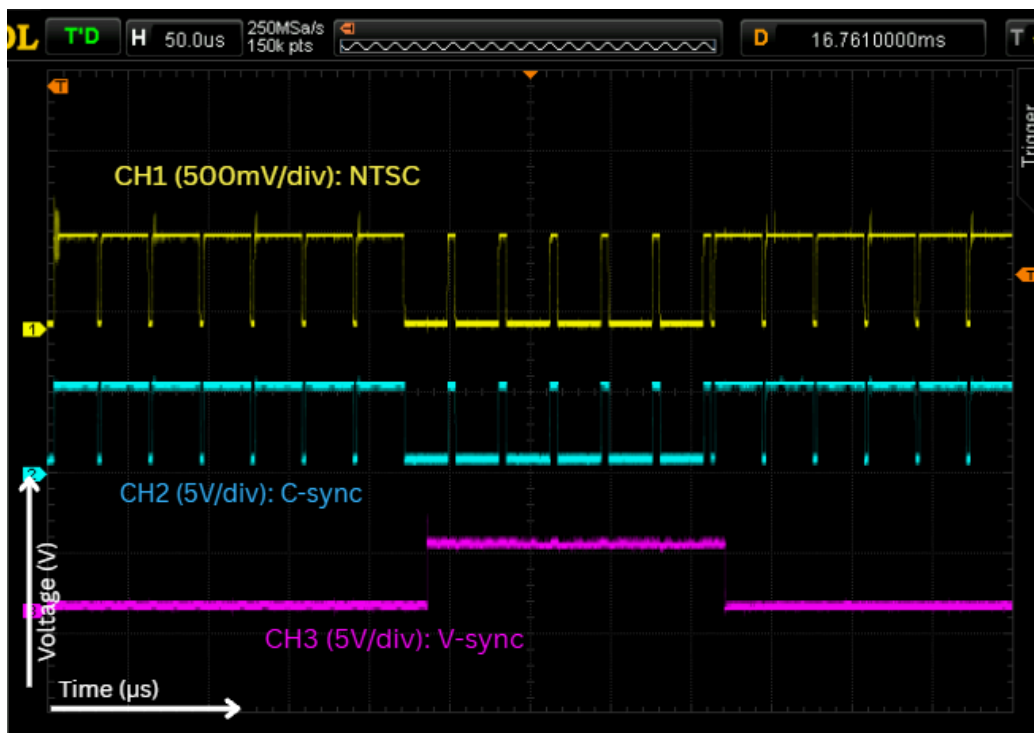


Figure 24: V-sync and C-sync Oscilloscope Capture

Interpretation of Results

In the first image, the low-pass filter successfully integrates the longer vertical sync pulses into a distinct downward voltage ramp while leaving the narrow horizontal pulses as minor high-frequency ripples. In the second image, the 555 timer's hysteresis successfully ignores those high-frequency ripples and cleanly triggers on the integrated vertical pulse, generating a crisp, square vertical sync pulse for the duration of the retrace interval.

Horizontal and Vertical Sweep Generators

Theory of Operation

The horizontal and vertical ramp generators work by using the sync separator pulses to periodically discharge a capacitor that is continuously charged by a constant current source, producing a linear voltage ramp. When the subcircuit receives a LOW output(0V) from the H-sync or V-sync separator the DC offset will pull the voltage to 7.2V, but since threshold level is at $\frac{2}{3} V_{CC}$ (8V) the top internal comparator will send a LOW signal $R=0$ keeping its previous state. This allows the capacitor to charge up, even though the capacitor is connected to DISCHARGE and TRIG pins, the internal transistor never turns on unless the input from the H-sync or V-sync separator surpasses the threshold level. When the subcircuit receives a HIGH output (5V) from the H-sync or V-sync separator the DC offset will pull the voltage to 12.2V surpassing the threshold level and turning on the transistor and dissipating the voltage in the capacitor until the capacitor reaches below $\frac{1}{3} V_{CC}$ (4V) causing the internal bottom comparator to turn off the transistor then causing the capacitor to recharge, repeating the cycle.

Measurement Procedure

1. Apply the NTSC video signal and observe the H-ramp and V-ramp outputs on the oscilloscope.
2. Compare each ramp with its corresponding sync signal to verify that the ramp resets at the correct time.
3. Measure the ramp frequency, peak-to-peak amplitude, and reset time for both the horizontal and vertical ramps.
4. Check ramp linearity by measuring several points along the active portion of each ramp.
5. Connect the H-ramp and V-ramp to the oscilloscope X and Y inputs and verify that a stable raster is produced.

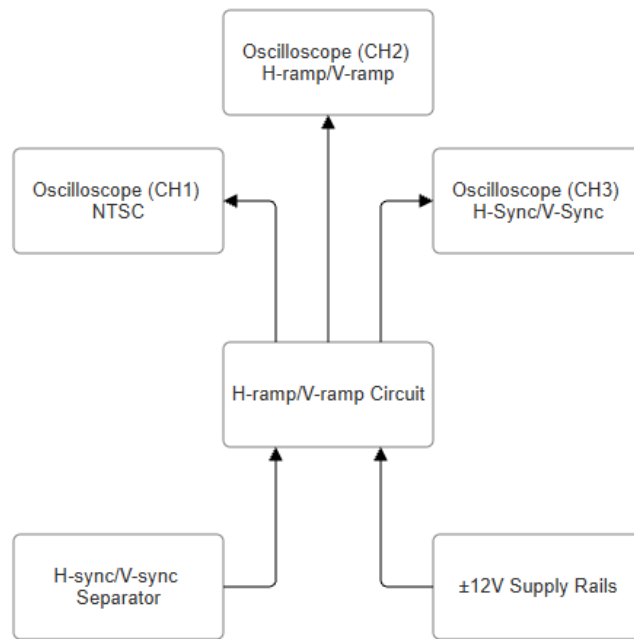


Figure 25: H-ramp and V-ramp Block Diagram Connections

Measurement Results

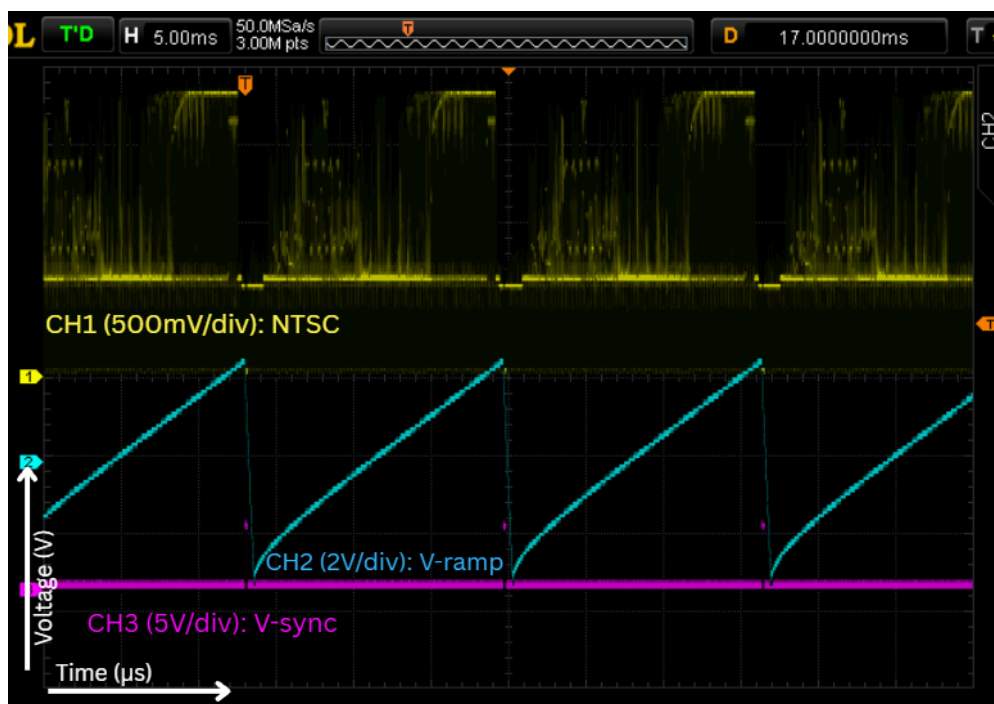


Figure 26: V-sync and V-ramp Oscilloscope Capture

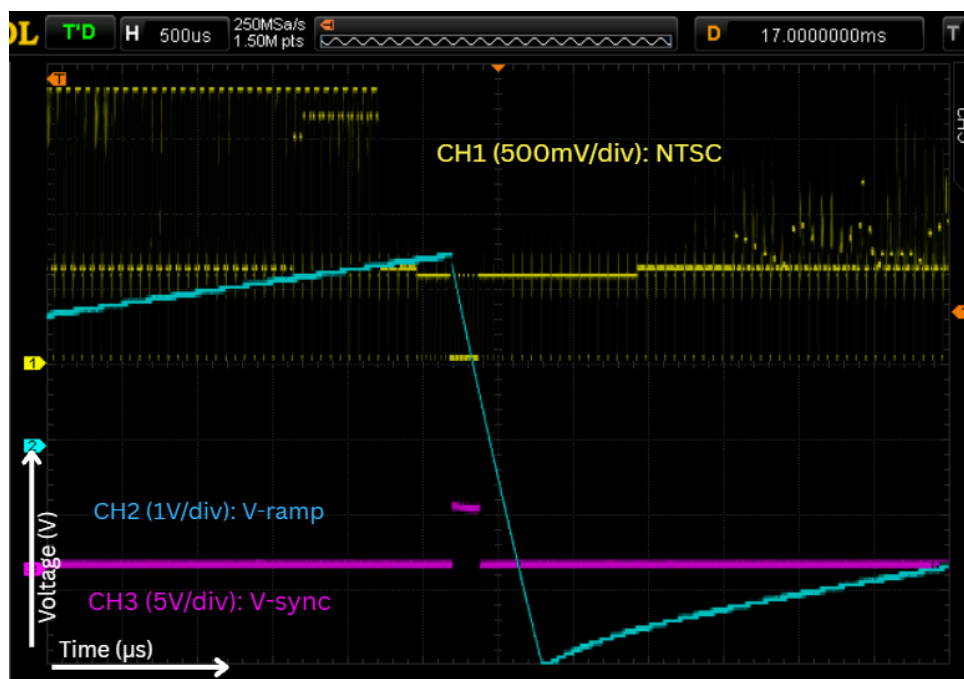


Figure 27: Zoomed V-sync and V-ramp Oscilloscope Capture

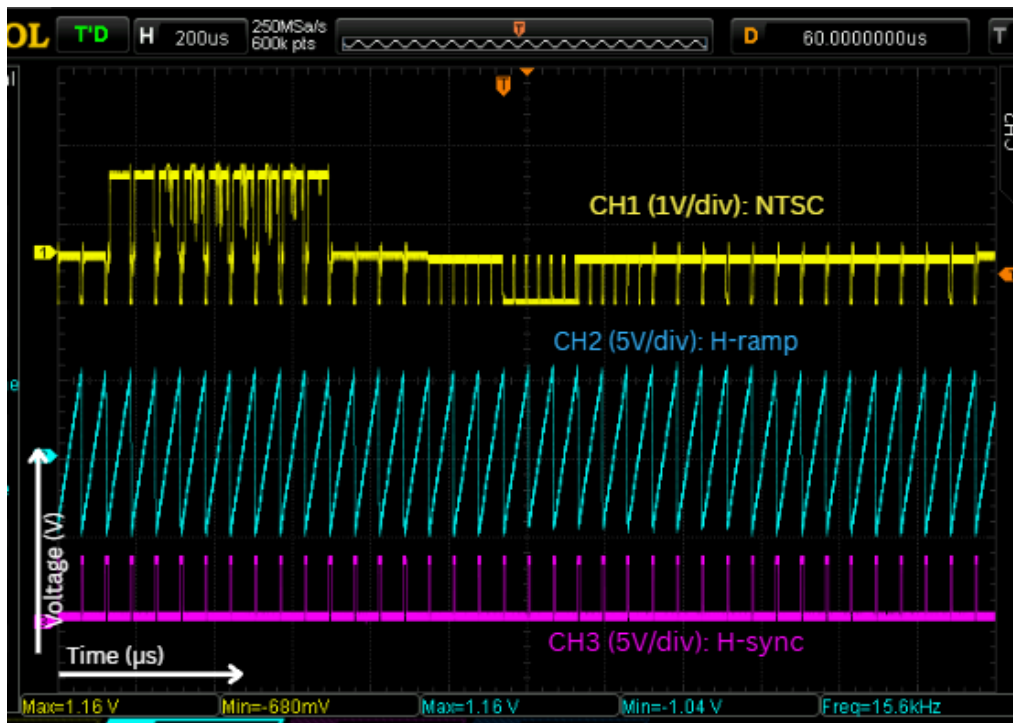


Figure 28: H-ramp and H-sync Oscilloscope Capture

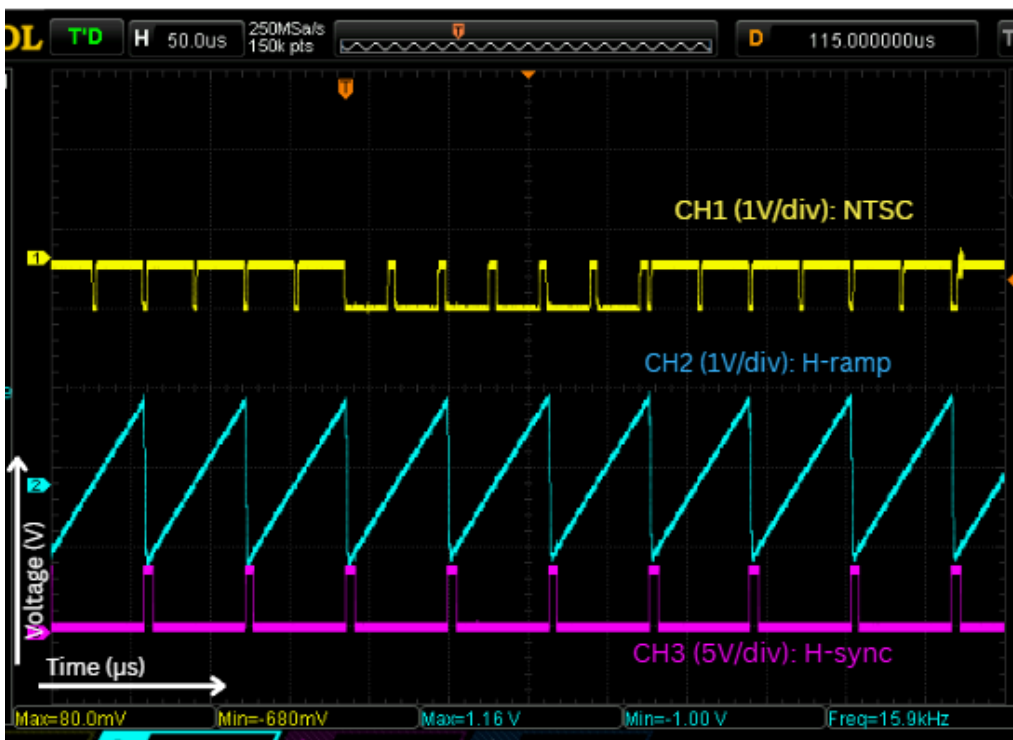


Figure 29: Zoomed H-ramp and H-sync Oscilloscope Capture

Interpretation of Results

The sweep generation subcircuits demonstrate clean linear ramp outputs for both horizontal and vertical deflection. The horizontal ramp generator outputs a $\sim 63.5 \mu\text{s}$ waveform at the line rate and the vertical ramp generator operating at a $\sim 16.67 \text{ ms}$ period (60 Hz field rate), each discharging smoothly upon receiving its respective extracted sync pulse (CH3) with minimal delay to ensure stable frame synchronization.

Blanking Generator

Disclaimer: The blanking generator circuit was not implemented in the final hardware system. Consequently, measurement results and result interpretations are omitted. The procedure below outlines the intended verification steps for the subcircuit.

Theory of Operation

The blanking generator uses the horizontal and vertical synchronization signals to generate a blanking flag during the retrace portions of the NTSC signal. The horizontal blanking signal remains active throughout the horizontal front porch, sync tip, and back porch, while the vertical blanking signal covers the corresponding vertical retrace interval. These signals are combined using an OR function so that the blanking flag is active whenever either horizontal or vertical retrace occurs. The blanking flag is then provided to the luminance image-processing stage, where it forces the Z-axis signal to the specified black level during retrace and prevents unwanted retrace lines from appearing on the CRT.

Measurement Procedure

The blanking generator was not successfully implemented in the final system. Therefore, the following procedure describes how the subcircuit would have been tested if it were functional.

1. Display the H-sync, V-sync, and blanking output signals on the oscilloscope.
2. Verify that the blanking flag is active throughout both the horizontal and vertical retrace intervals.
3. Connect the blanking output to the luminance-processing stage and verify that the Z-axis output is forced to the specified black level during retrace.

Luminance Separator

Theory of Operation

The luminance separator isolates the active video intensity information from the composite NTSC signal by stripping away synchronization pulses and color subcarrier interference by using a low pass filter with a cutoff frequency at 1.59 MHz to attenuate the 3.58 MHz chrominance

subcarrier and a clipping stage to truncate all negative-going sync tips below the black-level threshold, leaving only active luminance content.

Measurement Procedure

1. Apply the NTSC video signal and observe the clamped NTSC waveform together with the luminance separator output on the oscilloscope.
2. Verify that the sync portions of the waveform are removed or clipped while the active luminance information remains visible.
3. Change the image brightness and check that the luminance waveform still varies normally without the sync pulses reappearing at the output.
4. Measure the response of the luminance path using a sharp video transition or suitable test signal to estimate its bandwidth and confirm that it is sufficient to preserve the image detail.

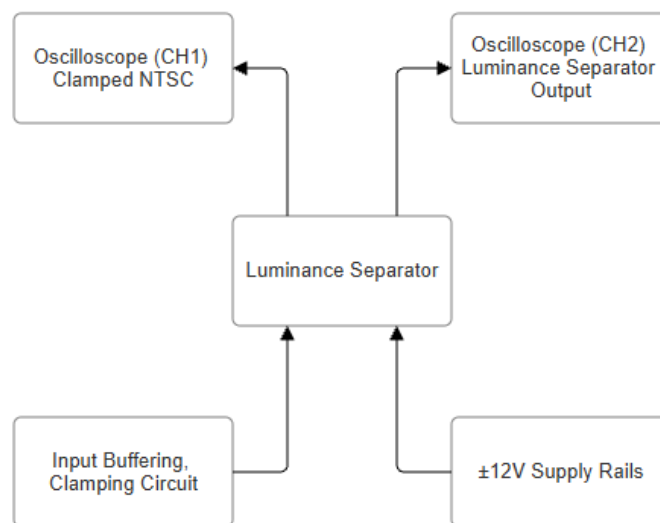


Figure 30: Luminance Separator Block Diagram Connections

Measurement Results

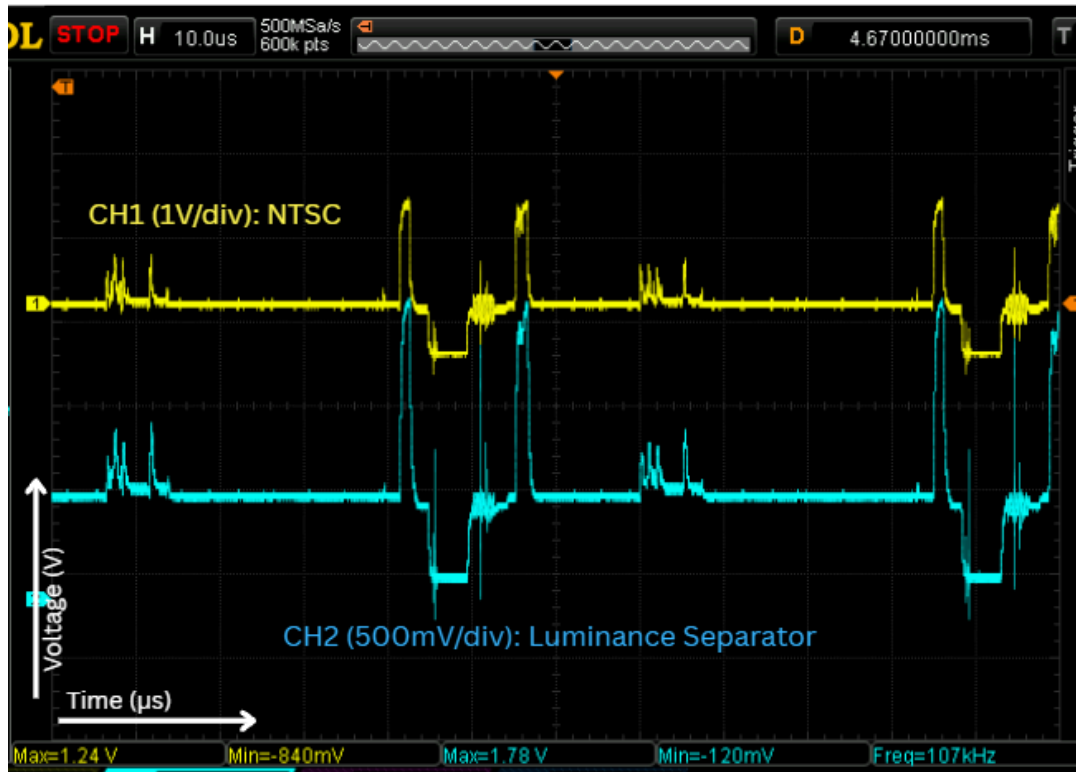


Figure 31: Luminance Separator Oscilloscope Capture

Interpretation of Results

The oscilloscope capture shows that the luminance separator is operating as intended by removing most of the negative-going synchronization content from the original NTSC signal while preserving the changing voltage levels that represent the video luminance information. Comparing CH1 and CH2, the large sync pulses present in the NTSC waveform are clipped to a nearly constant level at the separator output, while the smaller variations associated with image brightness remain visible. Some residual spikes and high-frequency components are still present due to non-ideal clipping and filtering, but the resulting waveform is suitable for use by the following brightness and contrast stages.

Luminance Image Processing

Theory of Operation

The luminance image processing works by scaling active video signal amplitude and setting the black-level offset using an op-amp configured as an inverting amplifier. The input video signal is fed through a $1\text{k}\Omega$ resistor in series with a $10\text{k}\Omega$ potentiometer which controls contrast into the inverting terminal, where a $100\text{k}\Omega$ feedback resistor establishes an adjustable inverting voltage gain range of -9.1V to -100V . A second $10\text{k}\Omega$ potentiometer which controls brightness is connected to the output of the amplifier and biased between 12V and ground through a $10\text{k}\Omega$. Creating a DC offset range of 6V to 12V . The contrast and brightness potentiometers operate independently because the contrast pot varies the input resistance to scale only the AC signal gain, while the brightness pot injects a variable DC bias into the summing node to shift only the DC baseline voltage.

Measurement Procedure

1. Apply an NTSC test image with several brightness levels and observe the processed luminance output at the oscilloscope Z-input.
2. Adjust the contrast control through its range and measure the change in luminance amplitude, checking that the blanking/black level remains approximately unchanged.
3. Adjust the brightness control and observe the resulting shift in output level while checking that the luminance amplitude is not significantly altered.
4. Measure several input and output luminance levels at selected control settings to verify that the processing stage remains approximately linear and to identify any clipping at the limits.
5. Connect the final luminance signal to the oscilloscope Z-axis with the X- and Y-ramps active, then verify that brightness and contrast controls produce the expected visual changes in the displayed image.
6. Observe the Z-axis signal during horizontal and vertical retrace and confirm that the blanking signal forces the output to the specified black level regardless of the brightness and contrast settings.

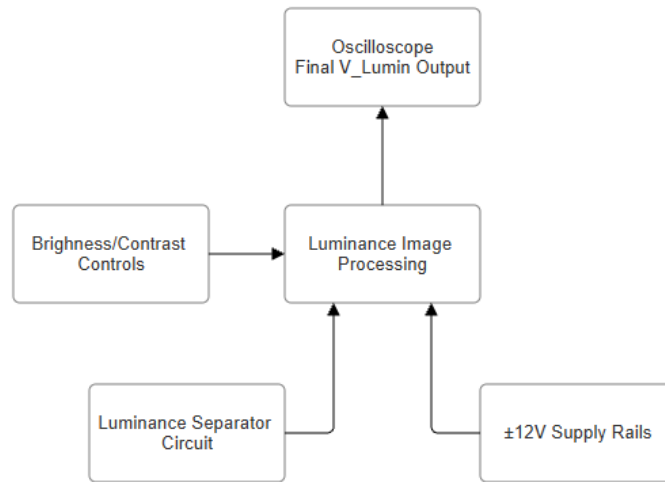


Figure 32: Luminance Image Processing Block Diagram Connections

Measurement Results

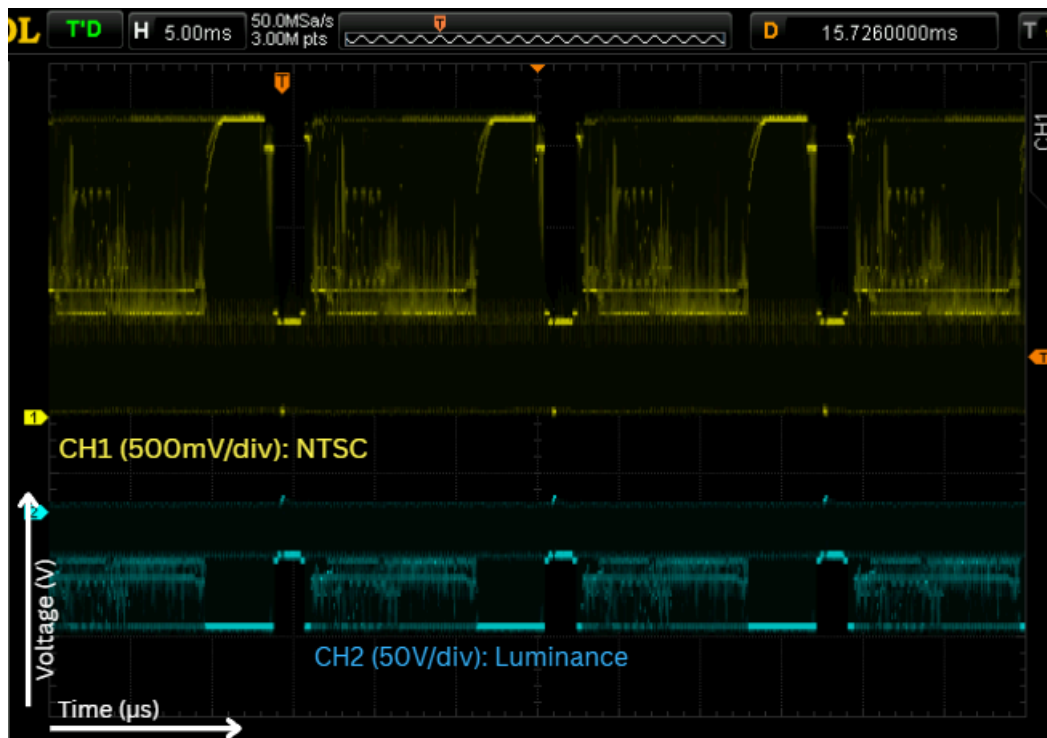


Figure 33: Luminance Image Processing Oscilloscope Capture

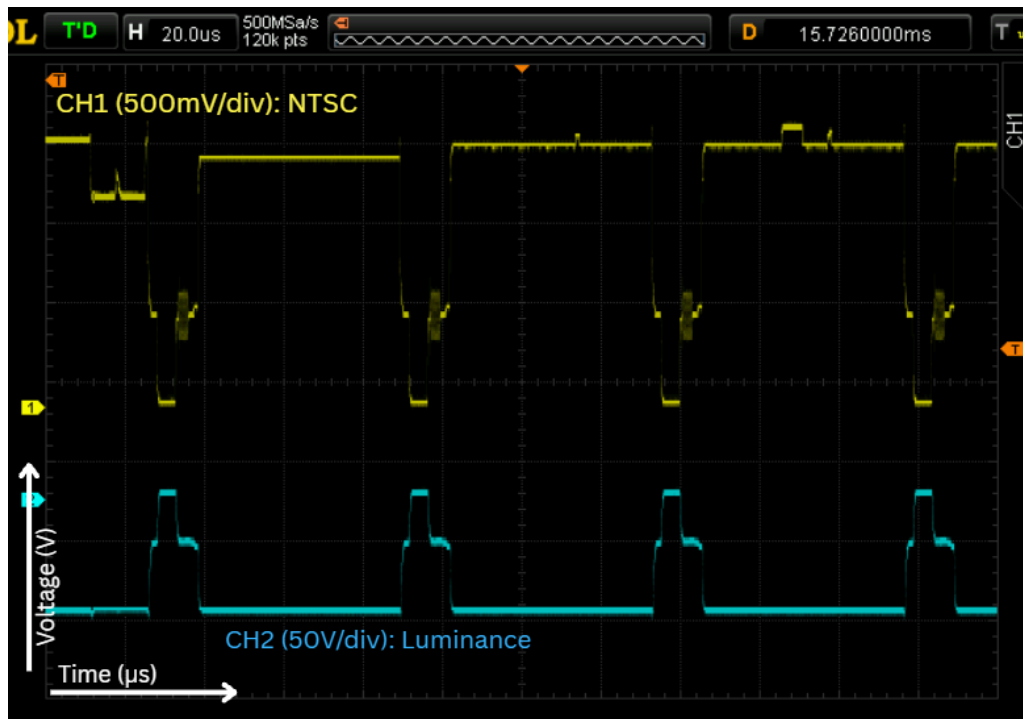


Figure 34: Zoomed Luminance Image Processing Oscilloscope Capture

Interpretation of Results

The oscilloscope captures show that the luminance image processing stage successfully conditions the separated luminance signal into a form suitable for driving the oscilloscope Z-axis. Compared with the original NTSC waveform, the processed CH2 signal has a shifted DC level and a larger, more clearly defined amplitude corresponding to the video brightness information. The zoomed capture also shows that the processed luminance follows the active-video portions of the NTSC signal while remaining relatively constant during the non-image intervals. Overall, the stage provides the expected brightness and contrast adjustment while preserving the main luminance variations needed to reproduce the image

Conclusion

In this project, an analog video decoding system was designed, built, and tested to process a standard NTSC composite video signal and output the corresponding X-, Y-, and Z-axis signals required to display video raster images on an oscilloscope CRT. The system integrated input clamping and filtering, composite sync extraction, horizontal and vertical sync separation, sweep ramp generation, luminance separation, and inverting luminance image processing.

Key design objectives were successfully realized: a passive diode clamp restored a stable positive DC baseline near 0.021 V, while a 1.59 MHz low-pass filter with a 1 M Ω discharge path attenuated the 3.58 MHz color subcarrier to ensure clean reference levels and prevent node floating. For timing separation, an LM311 comparator extracted composite sync tips at a 176 mV threshold, while interlocked dual-stage 555 monostable multivibrators masked double-frequency equalization pulses (31.468 kHz) to output a continuous 15.734 kHz horizontal sync stream.

An RC integrator (15.9 kHz) combined with a 555 hysteresis stage cleanly extracted 60 Hz vertical sync pulses, while Wilson current sources driving 555 timers produced linear horizontal ($\sim 63.5 \mu\text{s}$) and vertical ($\sim 16.67 \text{ ms}$) sweep ramps with fast retrace times. Additionally, clipping below the black level and amplifying with an LM6171 high-speed op-amp enabled independent gain and DC offset adjustments for CRT Z-axis contrast and brightness control.

While core raster scanning and luminance processing were fully functional, hardware trade-offs remained, including the lack of a fully integrated dedicated blanking generator relying instead on standard luminance level control and minor DC baseline drift under dynamic live NTSC video due to changing Average Picture Levels (APL). Overall, the design successfully validated the principles of analog video timing separation, sweep synchronization, and high-frequency active signal processing, with future improvements pointing toward an active DC restoration clamp and dedicated logic gates for complete retrace blanking.

Circuit Schematics

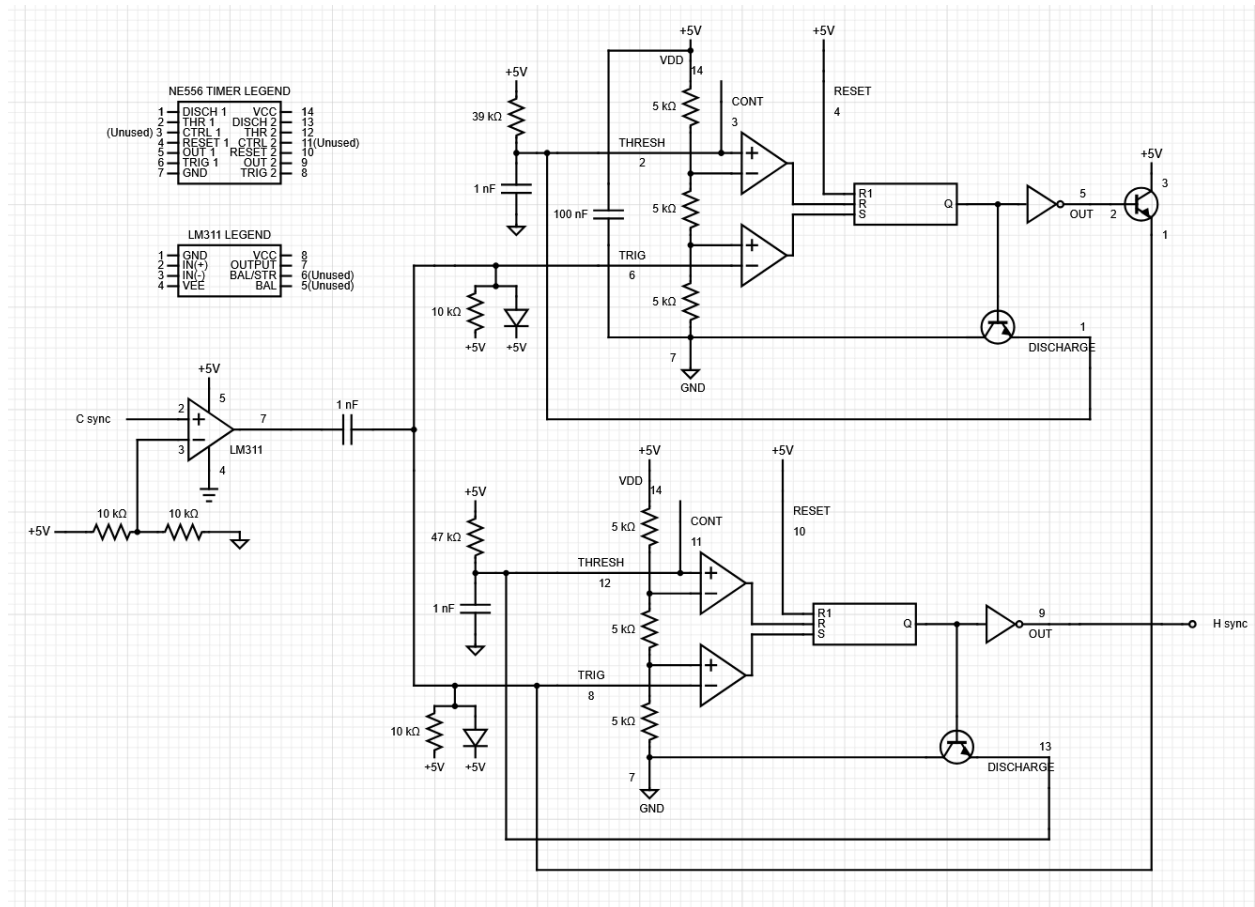


Figure 35: H-sync Circuit Diagram (3rd iteration)

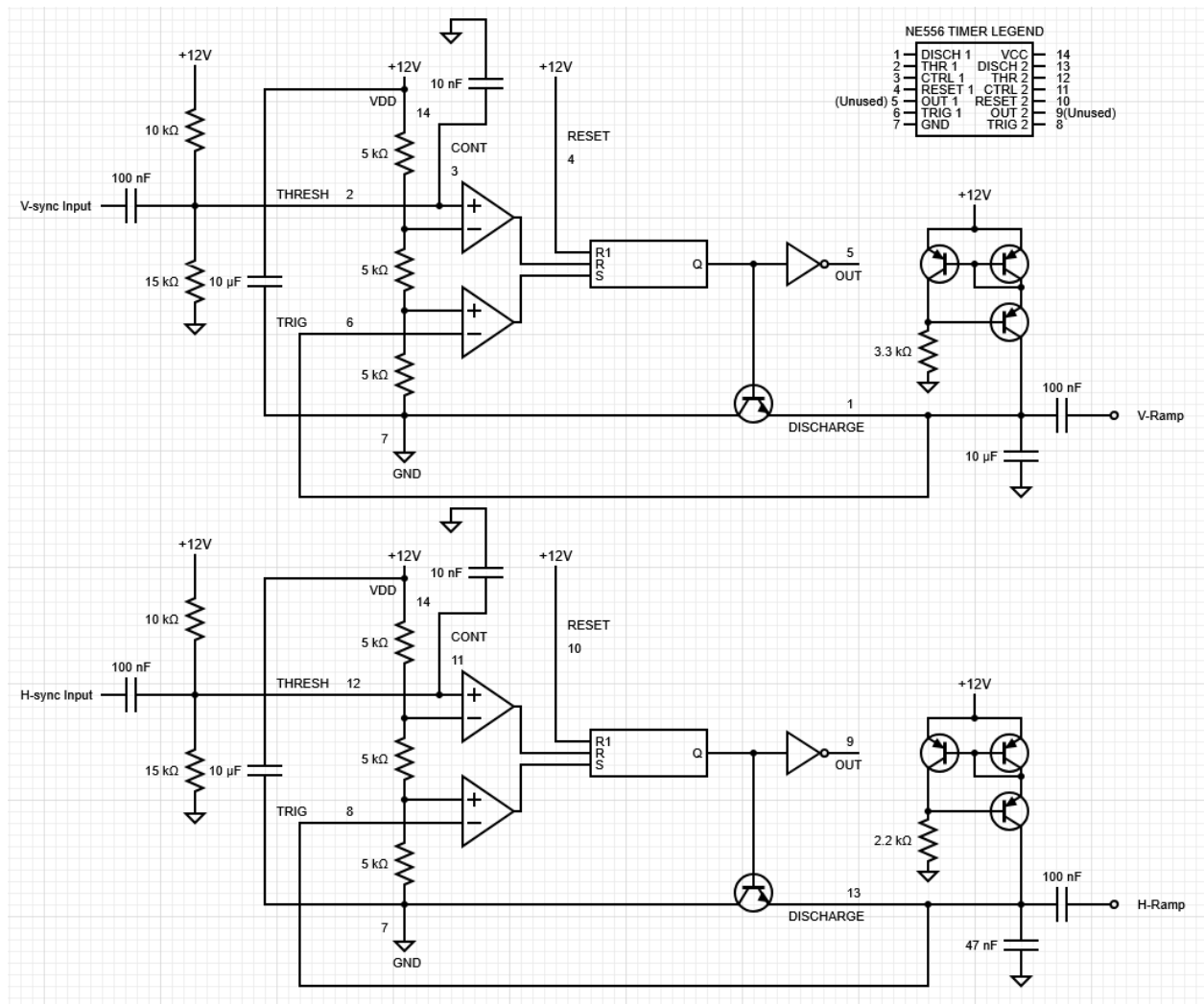


Figure 36: H-ramp and V-ramp (2nd Iteration)



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